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Music emotion recognition with cEEGrid

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E-mail: marcus.pearce@qmul.ac.uk**Keywords:** EEG, emotion recognition, mobile EEG, ear-EEG, deep learning, overfitting, data splitting

Abstract

Objective. Emotion decoding is a growing field of electroencephalography (EEG) research with numerous applications in areas such as healthcare, particularly when coupled with mobile EEG. However, research into mobile emotion decoding has been somewhat limited, especially regarding emotions elicited by music and how data splitting affects performance. The present study focuses on music emotion decoding with mobile (around-the-ear) EEG and the impact of the data splitting technique used. **Approach.** We collected the decoding auditory attention and musical emotions with ear-EEG dataset (DAAMEE), featuring both scalp EEG data (DAAMEE-s), and around-the-ear cEEGrid data (DAAMEE-c). DAAMEE featured several tasks, in one of which subjects listened to music and recorded their emotional state. We used the emotion task data, as well as the scalp EEG data from the existing DEAP dataset, for performance tests with various data splitting techniques and deep learning models. The tests focused on binary valence decoding, but the best-performing models were also applied for arousal, dominance, and three-dimensional valence-arousal-dominance classification. **Main results.** Performance with DAAMEE-c was generally similar to performance with DAAMEE-s and DEAP, showing the viability of cEEGrid for emotion decoding. The results also demonstrate that using certain data splitting techniques may lead to performance inflation due to mechanisms such as temporal correlations between samples being exploited. **Significance.** While these results motivate further research on mobile emotion decoding, future studies (for emotion decoding, and EEG classification generally), should take care to choose a data splitting paradigm that avoids overfitting.

1. Introduction

Historically, electroencephalography (EEG) has proven to be a valuable tool in both neuroscience research and medical practice. It has been used to study sleep [1, 2], consciousness [3], and neurological disorders [1–3]. It has also been used to diagnose and help manage various disorders [4–6]. Most of this work has taken place in laboratory or clinical settings, and analysis has largely been conducted manually or via relatively simple computational techniques such as time-frequency analysis [1–6]. However, in recent years, research has moved toward more progressive applications including emotion decoding [2], and the emerging subfield of wearable EEG technology is opening the door to many applications that

were once infeasible. Machine learning techniques for EEG analysis are an active and promising research area [2, 3].

The present work investigates the use of mobile EEG techniques (ear-EEG) for emotion recognition or decoding. While previous research using ear-EEG for emotion decoding has focused on short everyday sounds [7], pictures [8] and videos [9–15], the present research examines the feasibility of ear-EEG emotion decoding for music heard in isolation. We have followed a deep learning approach to analysis, with an examination of the impact of the data splitting technique used during training and testing. We also present a new dataset for scalp and ear-EEG analysis, which is, to our knowledge, the first publicly available ear-EEG dataset for affect recognition.

The paper is structured as follows. Section 2 reviews related work. The methods are presented in sections 3 (the datasets used and data collection), and 4 (preprocessing, machine learning pipelines, and evaluation). Section 5 reports the results, which section 6 discusses, while section 7 presents the conclusions. Appendices A–D contain additional information on the analysis and the models' learning behavior. Links to the original dataset, and to resources such as the analysis code, are given in the data availability statement and supplementary materials sections at the end of the paper. To avoid ambiguity, the term 'pipeline' will be used to describe a particular machine learning process (data formatting and additional preprocessing, feature extraction, and training/validation/testing), when working with a particular dataset that is split using a particular data splitting technique. This term will also be used when discussing machine learning processes in a general sense (for example, when discussing trends in the literature).

2. Related work

2.1. Emotion recognition

Emotion recognition, or emotion decoding, refers to the use of biosensory data to classify emotions. In the context of EEG, it involves using aspects of the neural signal (e.g. time-frequency distribution) to classify different emotional states, as a more direct measure than modalities that do not directly measure brain activity (e.g. electrocardiogram) [16–18]. This has applications in a variety of aspects of medicine: mental health research [19]; treatment of mental health conditions [20–22]; diagnostics, including for psychophysiological diseases [23]; and palliative care [24]. Furthermore, emotion decoding techniques may benefit technologies, such as robots, that are used to support those with disabilities [25, 26].

Work has focused on decoding emotions categorically ('joy', 'sadness', etc), and dimensionally, using continuous orthogonal scales to represent different aspects of emotion. The present work uses the valence-arousal-dominance (VAD) model, in which emotions are scaled from low to high for valence (negative/positive), arousal (passive/active), and dominance (submissive/dominant) [27]. Figure 1 illustrates how this corresponds to emotional categories. Various EEG feature extraction and machine learning methods have been tested for emotion recognition. Examples of relevant features include: spectral features such as power spectral density (PSD) [7]; temporal features such as Hjorth parameters [29]; spatial features such as common spatial pattern [30]; and measures blending spectral, temporal, and

spatial information, such as differential entropy (DE) [7, 31], entropy asymmetry [7], and topographic maps of spectral energy [7, 25]. Additionally, some studies have focused on end-to-end classification. In other words, feeding raw data and allowing the model to extract features [32–34], sometimes without prescribing a particular feature to focus on [35, 36]. Machine learning architectures that have been investigated include: shallow classifiers such as support vector machines and random forests [37]; and deep classifiers such as convolutional neural networks (CNNs) [25, 33], recurrent neural networks (RNNs) [38], graphical neural networks (GNNs) [39], and Transformers [40, 41]. Approaches involving hybrid [42, 43] and ensemble [44, 45] classifiers are increasingly prevalent. For comparisons of these feature extraction and machine learning methods see the reviews by Hamzah and Abdalla [46], Wang *et al* [47], and Lim *et al* [18].

2.2. Mobile emotion recognition

In recent years, there has been increased interest in developing mobile EEG technology, which can be used while a person goes about their daily life. In principle, mobile EEG should be unencumbering, discreet, comfortable, and simple to self-operate. A particularly versatile technique that is emerging is ear-EEG, in which signals are measured from inside or around the ears [48]. Spatial coverage is limited, and signals measured through ear-EEG are weaker than those measured through scalp EEG; however, the coverage of the temporal brain regions is still reasonable, and ear-EEG is more robust to ocular, motion, and electromagnetic noise [48]. Thus, for signals such as auditory event related potentials, the SNR can be on-par with those for scalp EEG [49]. Ear-EEG has shown promise in many applications, including: measuring hearing impairment and monitoring driver fatigue [48]; facilitating visual BCIs [50]; and emotion decoding [7, 8, 13]. It has been shown that combining ear-EEG with other modalities may be particularly effective for mobile assessment of emotions [9], especially stress [51, 52]. Many of the applications of emotion recognition, discussed in section 2.1, focus on helping patients and users with disabilities to regain independence, dignity, and comfort [20, 21, 24, 26], so emotion classifiers based on wearable EEG are particularly relevant. In recent years, Mai *et al* have shown that ear-EEG emotion decoding may be useful beyond the lab, by developing an ear-EEG device which decodes emotion in real-time and communicates with the user via a smartphone device [10], and by demonstrating a system which utilizes the Internet of Things to facilitate a range of applications [11]. Consumer mobile EEG technology may benefit society in general [53].

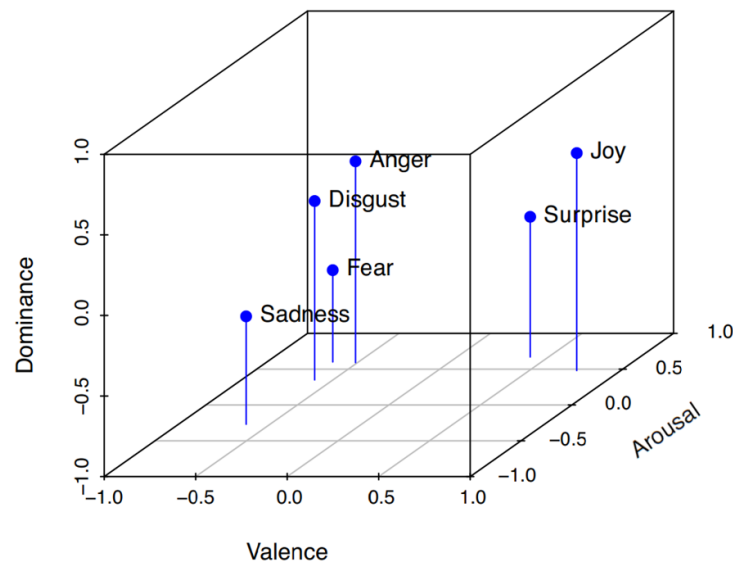


Figure 1. The mapping between the valence-arousal-dominance (VAD) model and the Ekman categorical model. Used under the terms of the [CC BY 4.0](#) licence [28]. Reproduced with permission from [28]. [CC BY-NC 4.0](#).

The studies cited above have investigated emotion decoding with ear-EEG in experiments involving stress-inducing tasks [51, 52], internet browsing or reading comments [12] and watching videos [9–12, 14, 15]. While one study focused on short audio clips of everyday sounds lasting 1.5–6 s [7] and another on three emotional videos paired with music [13], no study has examined emotion decoding from ear-EEG during pure unaccompanied musical listening, which was the goal of the present research. We use cEEGrid, where an array of electrodes is placed around each ear in a ‘C’ shape, as shown in figure 2. A typical cEEGrid setup consists of ten electrodes per ear, which may include the ground and reference electrodes [54]. Simpler and alternative configurations have also been explored [7, 55].

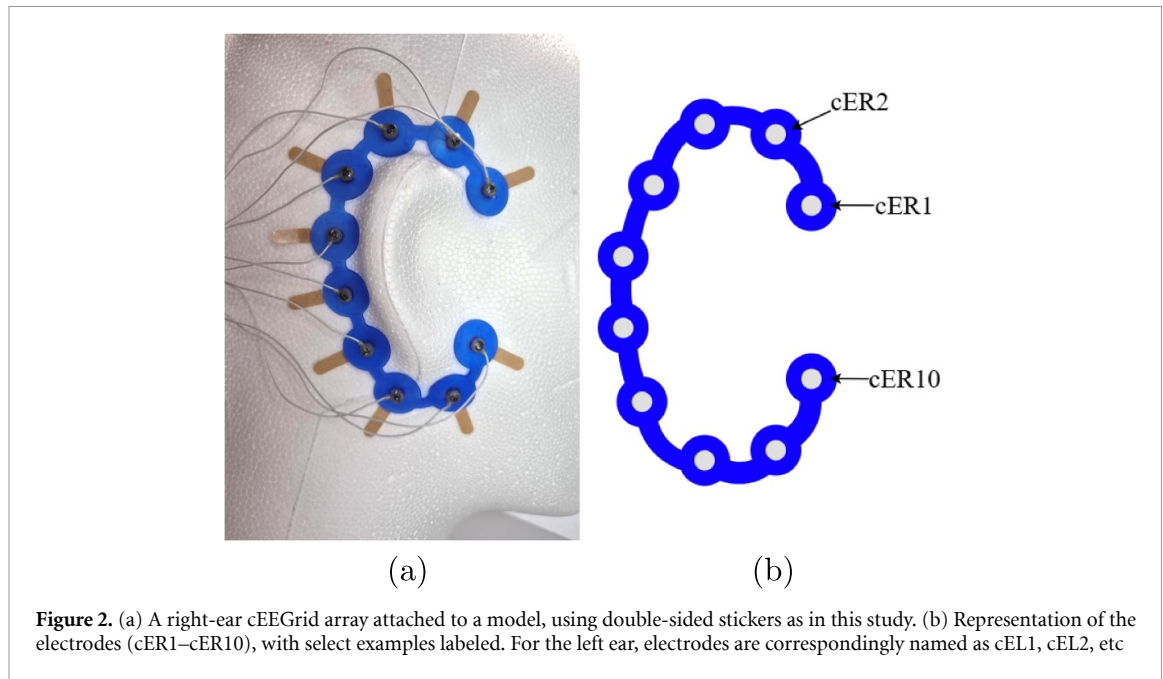
2.3. Data splitting

It is vital that any supervised machine learning pipeline is designed in such a way that data used for testing do not influence model training. Such influence is known as data leakage, or data contamination, and it can cause models to memorize features specific to the test data, allowing them to perform well during testing but generalize poorly when applied elsewhere. The most direct form of data leakage is when the same samples are used in both the training and the test data subsets [56]. Although it is crucial to avoid using this, there are other factors that can bias classifiers and lead to inflated test performance. The present work focuses on two of these.

The first issue to consider is overfitting caused by temporal correlations [57, 58]. The brain generates resting-state oscillatory EEG activity, especially in the

alpha and beta frequency bands, showing autocorrelation whose strength declines over time [58]. This means that even with no task, EEG samples will show greater correlation the closer together their temporal positions in the original recording. This creates a confound in block design experiments in which samples with a given label are temporally contiguous or close in time, such that machine learning algorithms can learn these temporal correlations distinguishing labels in the training set and thereby achieve artificially high performance on the test set [57].

In the context of single-subject emotion decoding, White and Power [59] tested three approaches, with a 5 s period considered as one sample for training or testing. The first approach was a randomized k -fold cross-validation (CV) with either 15 s or 60 s epochs being extracted from the data before shuffling. The second approach was a block-wise approach, in which epochs were divided in the same way, but were kept together: an entire trial with a given stimulus for a given subject would be fully included in either train or test data, meaning that temporal correlations between train and test data would not be present. The third approach, which they called the ‘ground-truth’ performance, involved splitting trials into 5 s (one sample) periods, with random shuffling: this would ensure no temporal correlations between train/test epochs, while still exposing the model to data from all trials. They found that the k -fold approach could inflate accuracy by as much as 25 percentage points relative to the ground-truth approach. Additionally, the block-wise approach caused accuracy to be 11 percentage points lower than the ground-truth approach, i.e. underfitting. This is because the block-wise approach only



included data from certain trials for training (limiting generalization), and because within-trial temporal correlations could still be learned, but would confuse a model, as opposed to helping it [59]. Similar trends have been shown for viewed object classification [57], mental workload classification [60], EEG-speech decoding [61], and motor imagery tasks [62].

Secondly, it has been shown that an analogous bias can occur when within-subject training and classification is performed, with classifiers focusing on subject-specific properties unrelated to the problem at hand [63, 64]. Brookshire *et al* showed that the accuracy of a binary Alzheimer's detector could be inflated from 53% (with chance performance included in the 95% confidence interval) to 99.8%, when the data splitting technique was changed from leave-one-subject-out (LOSO) CV to a segment-holdout CV technique. The latter technique allowed data from the same subjects to be used in both training and testing, and, specifically, involved having an equal amount of test data per subject per fold. For seizure identification, the difference in accuracy between segment-holdout and LOSO classification pipelines was lower but still considerable (79.1% and 65.1%, respectively) [65]. EEG patterns are highly individualized, to the extent that 'EEG fingerprinting' to identify subjects is its own area of research [66]. As an example, a periodic activity, which usually dominates PSDs, can vary substantially between subjects [67]. In studies where multiple recording sessions are completed for each subject, session-specific properties may be a distinct source of data leakage. These include impedance differences in electrode connections [68], as well as factors such as how well the subject slept the night before [69]. However, for

many datasets, including those used in this work (see section 3.1), a single recording session is conducted for each subject. As such, session-specific properties may effectively be considered an extension of subject-based properties in this context.

A less direct form of data leakage, not a focus in this work but discussed here for completeness, is overfitting caused by stimulus-specific responses. This is a concern in EEG studies where correlations between EEG data and particular stimuli are directly considered, such as auditory attention decoding (AAD) [70], and EEG-to-text [71]. However, for emotion-decoding, where the stimuli themselves are not directly relevant, this is still a concern. Bendrich *et al* [72] completed emotion decoding work using several data splitting techniques, including: leave-one-movie-out (LOMO), in which data relating to a given stimulus are used for testing, but excluded from training data, for a given loop; and leave-one-person-and-movie-out (LOPMO), in which a particular trial is used for testing and all data relating to the stimulus and/or subject are excluded from training. The results showed that LOMO and LOPMO-based pipelines performed consistently worse than their LOSO counterparts. This would indicate that stimulus-based leakage is at least comparable to subject-based leakage, and potentially more serious [72]. Additionally, Moreno-Alcayde *et al* showed that stimulus-based data leakage can inflate accuracy by >20 percentage points for emotional annotation of DEAP [73]. Section 6.6 discusses how the present work could be built upon to study this.

Indirect data leakage is a concern in machine learning generally [56]. Similar trends have been observed in MRI classification [74], for example. In audio-based research, the term 'weak contamination'

has been coined [75] to emphasize that while this is distinct from ‘strong’ leakage (using the same data points for training and testing), it still leads to overfitting. In EEG research, some studies have carefully considered this (e.g. Mikkelsen *et al* [76]). However, in many studies, the problem has been overlooked altogether. It has been shown that while techniques such as random k -fold CV can allow this indirect leakage, inflating results compared to when more stringent methods such as LOSO are used, many studies have used the former such methods without consideration of these issues, likely distorting their results [57, 60, 63, 64]. In the context of neurological disorder classification, Brookshire *et al* noted that only 27% of reviewed studies from 2018 onward could be confirmed to have taken steps to avoid subject-based data leakage [65].

As data leakage is an inherent issue affecting all classification pipelines, two important considerations arise. Firstly, it is a concern that many previous EEG studies may have been affected by data leakage. Secondly, the problem cannot be avoided through any particular machine learning techniques or preprocessing. Instead, care must be taken during data splitting, and it is wise to consider the risk of data leakage when reviewing results or pipelines. Although these are beyond the scope of the present work, Kinahan *et al* reviewed other subtle forms of data leakage, such as when the entire dataset (including test data) is used to inform feature selection [62]. The experimental method for the present work, discussed later, was designed to avoid such issues.

2.4. Studying emotion decoding with TorchEEG

Machine learning pipelines can be used effectively to analyze EEG data, but the development process has many steps: data preprocessing, feature extraction, data splitting, model training/validation, and testing. To facilitate such work, in 2022 Zhang *et al* published the TorchEEG Python library. TorchEEG contains five modules: datasets, for dataset loading; transforms, for preprocessing, feature extraction, labeling, and data formatting; model_selection, for data splitting; models, featuring a variety of deep learning models for classification and data augmentation; and trainers, for training, testing, and fitting. The idea behind TorchEEG is to provide a single library with all of the tools necessary for EEG decoding using deep learning, with a particular focus on emotion recognition work.

The paper accompanying the TorchEEG library tested various aspects of it [77]. This included comparing the classification performance when different models, hyperparameters, feature extraction techniques, and data splitting techniques were used in classification tasks across a series of benchmark datasets. These tasks involved classification of valence,

and in some cases arousal. The benchmarks used several emotion decoding datasets, including the DEAP dataset [78], discussed in further detail below.

The most striking outcome of the study came from the data splitting experiment, where the KFold (KF), KFoldGroupbyTrial (KFGbT), KFoldCrossTrial (KFCT), and LOSO techniques, detailed in figure 3(a), were compared. It was found that pipelines using the former two techniques, which allow data from the same trials to be used for training and testing, yielded vastly higher performance accuracies compared to pipelines using the latter two, in which data from separate trials were used for training and testing. For example, the test accuracy of a binary continuous CNN (CCNN) [79] classifier for valence in DEAP was 93.20% when KF was used, but only 53.66% when KFCT was used. This indicates that the deep learning algorithms featured in TorchEEG (or at least, the CCNN) are particularly vulnerable to overfitting caused by temporal correlations as described in section 2.3, when care is not taken to partition the training and test data appropriately. Figure 3(b) illustrates how each of the data splitting techniques allows sharing of data from trials, subjects and stimuli. Another important result is that during model comparison tests, from a selection of CNNs, GNNs, and Transformers, the CCNN model was shown to generally achieve the highest performance [77].

2.5. The present work: mobile EEG emotion decoding with a focus on data splitting

The present research aimed to build on the previously discussed work on emotion decoding, mobile EEG, and data splitting techniques, utilizing the TorchEEG library. In particular, there were three aims: to validate the trends shown by Zhang *et al* with the DEAP dataset; to extend this with our original scalp data, which have similarities to DEAP with a focus on music rather than music video; and to expand further with our cEEGrid data, to investigate whether similar results can be obtained with around-the-ear EEG. The final aim is particularly important for the development of mobile emotion decoding systems which may one day be used in a range of applications beyond the laboratory. To facilitate comparisons with the results of Zhang *et al* and robustly address data leakage, we take a critical but open-minded approach when considering the results achieved with different data splitting techniques. There are some additional significant adaptations from the Zhang *et al* work [77]: a more expansive comparison of data splitting techniques to better investigate possible cross-subject data leakage; the exclusion of some models informed by existing results, but with the additional inclusion of an RNN model, to give a streamlined but more comprehensive analysis; consideration of dominance and eight-class VAD classification for a more comprehensive analysis

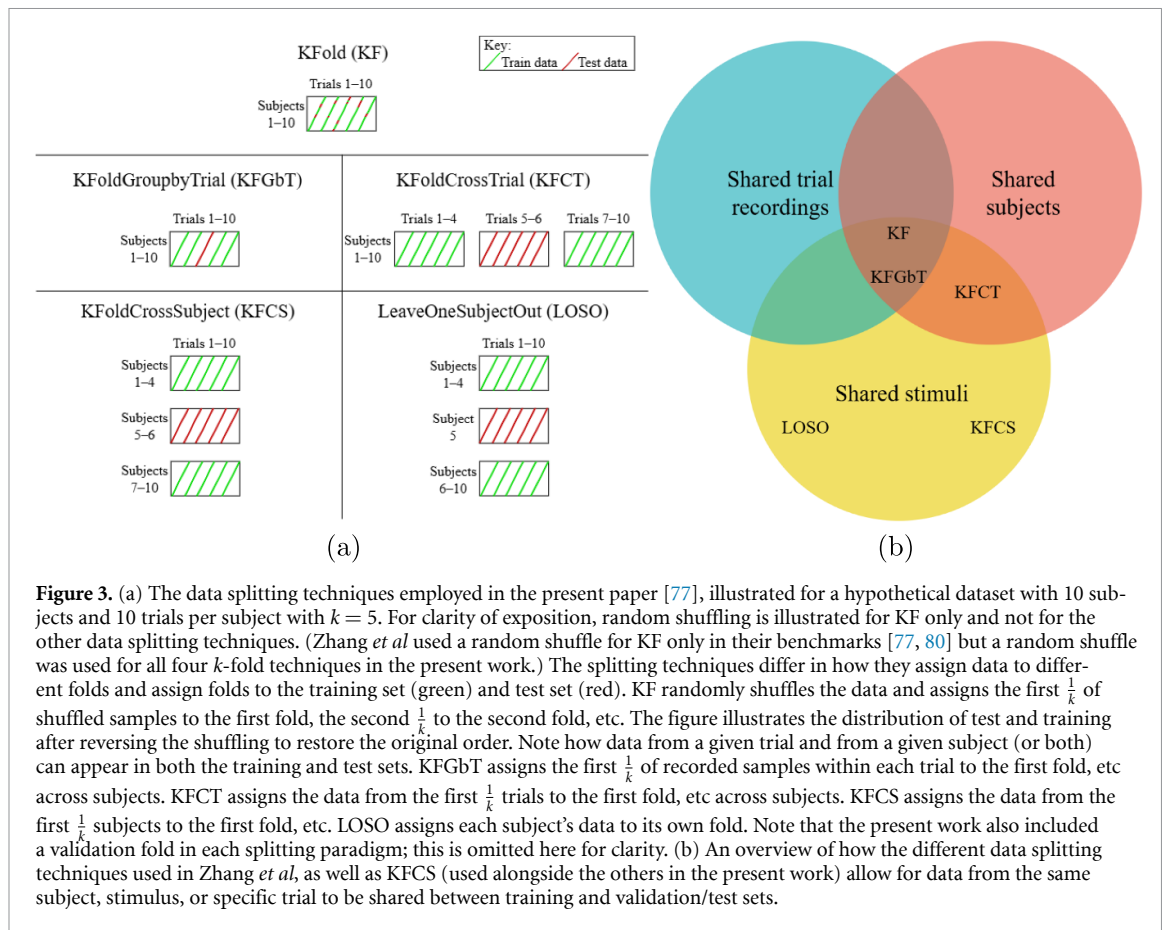


Figure 3. (a) The data splitting techniques employed in the present paper [77], illustrated for a hypothetical dataset with 10 subjects and 10 trials per subject with $k = 5$. For clarity of exposition, random shuffling is illustrated for KF only and not for the other data splitting techniques. (Zhang *et al* used a random shuffle for KF only in their benchmarks [77, 80] but a random shuffle was used for all four k -fold techniques in the present work.) The splitting techniques differ in how they assign data to different folds and assign folds to the training set (green) and test set (red). KF randomly shuffles the data and assigns the first $\frac{1}{k}$ of shuffled samples to the first fold, the second $\frac{1}{k}$ to the second fold, etc. The figure illustrates the distribution of test and training after reversing the shuffling to restore the original order. Note how data from a given trial and from a given subject (or both) can appear in both the training and test sets. KFGbT assigns the first $\frac{1}{k}$ of recorded samples within each trial to the first fold, etc across subjects. KFCT assigns the data from the first $\frac{1}{k}$ trials to the first fold, etc across subjects. KFCS assigns the data from the first $\frac{1}{k}$ subjects to the first fold, etc. LOSO assigns each subject's data to its own fold. Note that the present work also included a validation fold in each splitting paradigm; this is omitted here for clarity. (b) An overview of how the different data splitting techniques used in Zhang *et al*, as well as KFCS (used alongside the others in the present work) allow for data from the same subject, stimulus, or specific trial to be shared between training and validation/test sets.

of emotional states; and the use of significance testing to identify which pipelines significantly exceeded chance performance.

3. Methodology: Datasets and data collection

3.1. Datasets

Two datasets were used: the pre-existing DEAP dataset; and the novel 'decoding auditory attention and musical emotions with ear-EEG' dataset (DAAMEE, pronounced 'dame').

3.1.1. DEAP

DEAP has been used in many emotion recognition studies [18, 27, 46, 47, 59], and is the only dataset that is used throughout the reported tests of the TorchEEG library [77]. Thus, our results using DEAP can be directly considered against those in the literature. Additionally, DEAP bears several similarities to the DAAMEE dataset, an important one being that both include VAD emotion ratings. This makes it straightforward to compare the results from emotion decoding analyses of the two datasets.

DEAP features EEG recordings (alongside other modalities) taken whilst subjects watched music videos, as well as self-reported ratings including

for valence/arousal/dominance. Specifically, there are 40 trials per subject, each trial being 60 s long with a 3 s baseline, across 32 subjects. Emotion ratings were made on a continuous scale of 1.00–9.00 (binarized into low and high with a threshold of 5.00 in this work). The EEG data were collected at 512 Hz with 32 scalp channels located according to the 10–20 system [78]. We used the preprocessed version of DEAP provided online. This preprocessing contained the following steps: data were downsampled to 128 Hz; ocular artifacts were removed using electrooculography (EOG) data; a bandpass filter of 4–45 Hz was applied; the data were re-referenced to the common average; the channels were reordered for convenience; and finally, the data were segmented and ordered into 60 s trials, with the 3 s baseline being removed [81].

3.1.2. DAAMEE

DAAMEE is a new dataset collected for this research to study both emotion and attention decoding using ear-EEG (here we focus on the former). To our knowledge, there are no existing ear-EEG datasets for affect decoding. DAAMEE also contains scalp data, which is useful in itself for the present work, but also enables us to validate results from using cEEGrid recordings. Given that DAAMEE is a new dataset and the inherent

noise involved in EEG recording (see section 6.5.1), DEAP is used for benchmarking purposes.

DAAMEE consists of original musical stimuli (used throughout the experiment) and data from 31 subjects. The experiment contained five sections, with EEG being recorded during the final three: a hearing calibration test, a questionnaire, an emotion-decoding task, a multiple-stream AAD task, and a single-stream AAD task. Data from the AAD tasks are not considered in this paper. Data were collected simultaneously using scalp EEG (DAAMEE-s) and cEEGrid (DAAMEE-c). Data collection and preprocessing are discussed in detail in sections 3.2–4.1.

3.2. Subjects

Data were collected from 31 subjects (26 ± 9 years; 11 M, two trans. M, 18 F; three left-handed). An additional subject took part; however, technical issues discovered later meant that these data were not usable. It should also be noted that for two of the subjects cEEGrid data of sufficient quality was not collected, i.e. DAAMEE-c only features data from 29 of the subjects. For one subject, this was due to poor electrode connections; for the other, technical issues prolonged the experimental session, and the data amplifier ran out of charge before the main experiment. The experiment was conducted in English and all subjects were fluent English speakers. No subjects had a known history of neurological disorders. One subject had minor hearing loss, but it was decided that they would be able to participate as this has no effect on their daily life (they do not wear a hearing aid) and is primarily related to low-frequency sounds, which are not featured in the stimuli. Additionally, it was expected that any relevant hearing differences would be mitigated through the hearing calibration test early in the experiment.

3.3. Stimuli

Rather than using existing music, original musical stimuli were composed to ensure they would be unfamiliar to subjects and to allow us to impose several experimental constraints.

It was decided that stimuli should be 30 s long, the length of a trial; using longer stimuli which would end abruptly may have distracted subjects. To help account for entrainment [82, 83], a common tempo of 80BPM was chosen. The stimuli were created as three-part compositions ('sets'), with each composition containing a harmonica, vibraphone, and piano/keyboard part played to a chord progression which was consistent throughout that composition. This approach was followed to satisfy certain constraints in the AAD sections of the experiment, which used the same stimuli. One such constraint was that the instruments used must have similar attentional salience. In a related study [84], it was

found that this combination of instruments worked well during pilot studies³ This arrangement also has two benefits in the context of emotion decoding: firstly, using a variety of instruments may have benefited subject engagement; and secondly, using equally salient stimuli ensured fairness between trials. Within these constraints, four musicians composed twelve three-part compositions (a total of 36 stimuli), with each composition being intended to express a different emotional state within VAD space. Based on these expressed emotions, five compositions had high valence, six low valence and one neutral valence; five compositions had high arousal and seven low arousal; while four compositions had high dominance, seven low dominance and one neutral dominance.

3.3.1. Stimulus preprocessing

Stimuli were converted into monoaural tracks for convenience and out of necessity for the AAD parts of the experiment. Pieces were RMS normalized to -25.00 dB relative to full-scale deflection for consistency. Informed by a pilot study with nine subjects, the normalization levels for pieces in composition 'Set07' were adjusted to improve consistency with other stimuli in terms of perceived loudness. These were changed to -23.30 dB, -19.67 dB, and -23.15 dB for the vibraphone, harmonica, and keyboard/piano tracks respectively. After normalization, a fade-in and fade-out period of 0.6 s was applied to each stimulus to make the listening experience more comfortable, based on pilot study feedback.

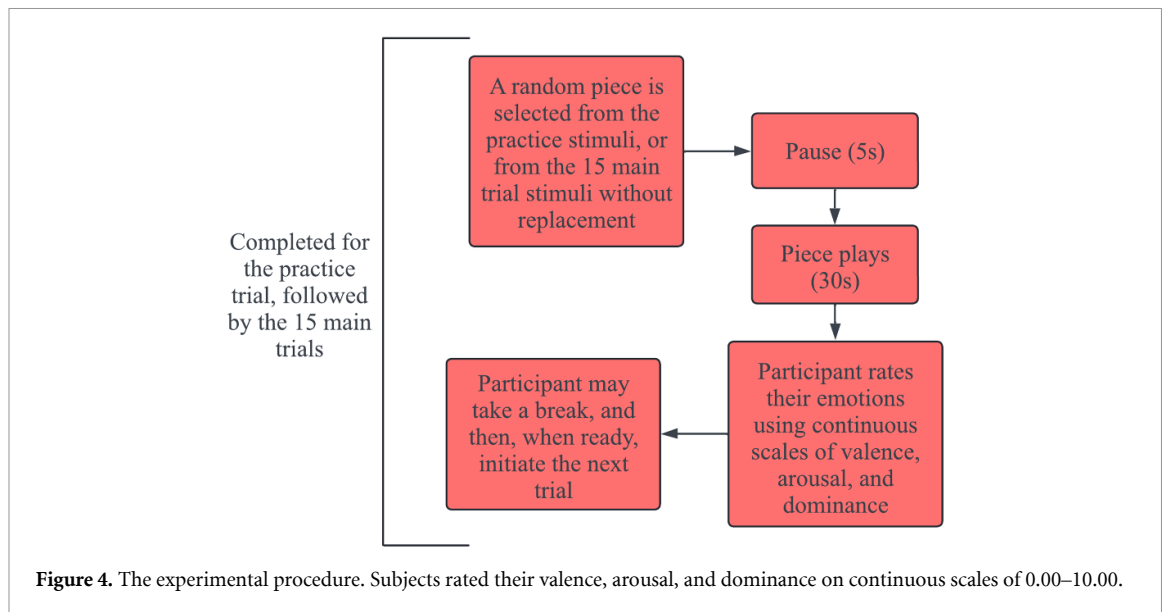
3.3.2. Stimulus division

The stimuli and subjects were both divided into two groups, with each group hearing a different half of the stimulus collection. The two stimulus groups were stratified to contain approximately equal numbers of compositions for each emotional class (e.g. 'high valence'). Similar stratification was applied regarding the musicians who produced the pieces since they each had slightly different styles, for example use of analog, digital, or virtual instruments. For consistency, each group was assigned a given practice composition from the stimuli, and each subject would hear a randomly chosen part from their group's practice composition. Consequently, each subject heard one practice piece which could be any instrument, and five main trial pieces for each of the three instruments (16 stimuli in total per subject).

3.4. Recording setup and experiment

For scalp recordings, the device used was a cap with 32 Ag/AgCl electrodes (placed according to the 10–20 system). For cEEGrid recordings, the device used

³ Advised by the lead author (W.W An, PhD, e-mail, March 2022).



had 21 IrO₂ electrodes (10 per ear, plus an Fpz electrode). A ground electrode was placed on the chin. For each device, EEG was recorded to a TMSi Mobita amplifier, with an average reference of all active channels. A sampling rate of 1 kHz was used throughout. Before application of the cEEGrid and ground electrodes, each subject's skin was cleaned using alcoholic wipes.

Subjects were asked to close their eyes just before the start of each trial. This was primarily to minimize EOG artifacts [1], but was also shown to benefit concentration during pilot studies. For smooth spatialization, to suppress external sounds, and to prevent any acoustic interference, audio was played through headphones. Throughout the experiment, messages on a screen were used to communicate instructions in a way that did not introduce unnecessary distractions. However, verbal instructions were given at the start of each task, and an experimenter was always present to advise. The experiment was completed via GUIs created through PsychoPy [85].

The experiment began with a hearing calibration test, with three sections (one of which was only relevant to AAD). In one section, subjects used a sliding scale to center the stereo balance of each instrument, to account for differences in perception, and in another they set the volume of each instrument to a comfortable level. Following the calibration test, each subject completed a brief questionnaire featuring: a demographics section; continuous VAD scales of 0.00–10.00 to rate current their emotional state; questions about their mental state (for example, tiredness); and the perception and emotion modules of the Goldsmiths Musical

Sophistication Index [86]. Given lack of agreement between existing studies on the response scales used to give VAD ratings [7, 78, 87, 88], 0.00–10.00 was chosen to provide intuitive minimum and maximum poles with unambiguous interpretations and a clear midpoint (5).

The emotion-based section of the experiment began immediately thereafter; figure 4 summarizes the procedure.

4. Methodology: analysis

4.1. EEG preprocessing

Appendix A gives details of the software used. Preceding analysis, the relevant standard 10–20 system channel coordinates were registered with DAAMEE-s. There were no existing standard cEEGrid location files, so an original montage was created by approximating two 'O' shapes around the ears, using the 10–20 system's A1 and A2 locations as reference points [89]. For each cEEGrid recording, the Fpz channel was removed (as it was not used for the analysis).

First, the data were referenced to the common average. A 1–45 Hz bandpass filter was applied to remove low-frequency drift noise [90], 50 Hz line noise, and other noise/brain signals outside the most relevant frequency range [7, 46, 68, 81, 87, 88]. A symmetric linear-phase (corrected to zero-phase) FIR filter with a Hamming window was used; this is the default filter used in MNE-Python 1.6.1, and gives a good balance between out-of-band suppression, reliability, and filter artifact minimization [90, 91].

Trials were extracted to 28.42 s epochs: although each trial was 30 s long, a 0.79 s buffer period was applied to account for the fade-in/fade-out periods of the stimuli (0.6 s) and eye opening/closing artifacts. Using 28.42 s epochs was convenient for splitting into sub-trial epochs during later analysis, as discussed in section 4.3. Linear detrending was applied during epoching to further reduce drift noise [92]. Next, z-normalization was applied within each epoch and channel to ensure efficient training [33, 93, 94]. When DEAP was used in the analysis, normalization was only used for certain model pipelines (see section 4.3), however, given that DAAMEE is a novel dataset with potential sources of noise (see section 6.5), it was considered best to normalize the data consistently throughout. Finally, data were downsampled to 200 Hz, in accordance with MNE-Python guidelines [90]. VAD ratings were binarized into ‘low’ (<5.0) and ‘high’ classes for each dimension.

4.2. Data splitting

Section 2.4 introduced some of the data splitting methods in TorchEEG, including those used in the benchmarks [77]. The present work largely followed this approach, but with three key differences. Firstly, to help avoid overfitting, a validation subset was featured in each data splitting paradigm, whereas to our knowledge Zhang *et al* simply used train/test splitting. For k -fold split techniques, they used $k = 5$, giving an 80:20 train:test ratio. We instead used $k = 10$ with a validation set, giving an 80:10:10 train:validation:test ratio. In the case of LOSO, in addition to using one subject’s data as the test set for each fold, a different subject’s data were used for validation. TorchEEG data splitting techniques are not currently designed to create validation sets, so this change required writing additional code. Secondly, the original study used LOSO to implement cross-subject training and testing. In LOSO, all of a subject’s data, and only their data, are used for the test subset (and a separate subject’s data are used for the validation subset if used, as in the present work), meaning that the results are vulnerable to variation between individuals. This is an instance of the bias-variance trade off in which a larger number of CV folds (with LOSO being the maximum) can reduce bias (due to increased training set similarity) but also increase variance due to increased (maximal) difference between test sets [95, 96]. Therefore, we additionally used the KFCS technique which includes more than one subject in each fold, thereby reducing variance between test sets (while also increasing bias). Lastly, whereas Zhang *et al* implemented random shuffling for KF splits, and not for KFGbT or KFCT splits, we use random shuffling for all three of these, and also KFCS.

The reasoning for this was that allowing the random distribution of stimuli in one splitting technique but not the others would produce unfair comparisons.

4.3. Machine learning

A selection of architectures was used to represent the major categories of neural network [97]: GNN, CNN, Transformer and RNN. Given that the CCNN was shown to generally outperform the other CNNs featured by Zhang *et al*, the others were excluded. Zhang *et al* did not include any RNN models, so as an expansion we decided to use one of the RNNs in TorchEEG. Of those available, the long-short term memory (LSTM) and gated recurrent unit models, neither has been shown to perform consistently better overall, but the LSTM is the more elaborate of the two and is generally better for dealing with longer and more complex data sequences [98, 99], making it appropriate for problems such as emotion decoding [26, 36, 100]. Thus, the LSTM was chosen as it was expected to be better-suited to the problem, and potentially more robust against noise (at least transient noise).

For several of the models, feature extraction included placing the electrodes into a grid based on their coordinates in the overhead plane. Given that for cEEGrid all electrodes are concentrated around the ears, it was not clear how this could be done in a meaningful way, so these models were omitted from the cEEGrid analysis. For the local-global-graph network (LGGNet) [101], a graphical network which requires electrodes to be grouped into regions, each cEEGrid electrode was assigned its own region for simplicity.

Given the large number of models considered, the model hyperparameters and feature extraction methods given in the TorchEEG documentation examples [80] were followed. In some cases, these hyperparameters are different from the default values. Appropriate adaptations were made for the DAAMEE dataset. An exception to this approach was made for the DEAP CCNN pipelines. Some of the documentation pages, [80], as well as comments in the source code, advise applying offline transforms to the baseline signal and subtracting this from the rest of the signal, for analysis of the preprocessed version of DEAP using the CCNN. However, as section 3.1 notes, the preprocessed version of DEAP already has this removed. Thus, we have excluded this from our DEAP CCNN pipelines. However, for completeness, and as it is not known which method was used in the reference study [77], valence classification was also run using the alternative approach; for convenience, this extension to the study will be referred to as DEAP-CCNN-Alt. Full details on this, the results, and the training epoch data are given in appendix D.

Appendix A gives details of all feature extraction methods and model hyperparameters (excluding for DEAP-CCNN-Alt). The analysis code (including for DEAP-CCNN-Alt) is linked in the supplementary materials section at the end of the paper.

As discussed in section 6.5.2, to maximize the amount of data for each subject, the practice trials were included in the analysis. Informed by the literature, Zhang *et al* advise using windows of 1–2 s for emotion decoding (see also [39, 102–104]). A 1 s window was used throughout. In preliminary testing, 1 s and 2 s windows were compared, but differences in performance were minimal. Thus, for comparability with previous results, we used non-overlapping 1 s sub-trial epochs when working with DEAP. For DAAMEE, sub-trial epochs of 1.015 s (203 samples) were used: using these sub-trial epochs, taken from trial epochs of 28.42 s, allowed us to ensure all 28 sub-trial epochs were of equal size whilst still having the desired buffer periods (see section 4.1).

Early stopping was implemented. Given the range of models and feature extraction methods used, predicting training dynamics was especially difficult. Thus, a conservative standard was followed: a minimum accepted ‘increase’ of 0.0% in validation accuracy, with a patience of 10 epochs (i.e. if validation accuracy decreased for 10 consecutive epochs, then training would be stopped). This is a compromise between a fairly aggressive setup, which might cause premature stopping in some situations, and no early stopping, leading to inefficiency and even overfitting elsewhere.

ClassifierTrainer, TorchEEG’s general-use trainer for classifiers, was chosen to ensure balanced performance. A learning rate and weight decay of 10^{-4} were chosen. Although the respective default values are 10^{-3} and 0.0, the chosen values are used in the documentation most frequently [80]. Additionally, it was considered that having a slower learning rate might benefit convergence, and having a non-zero weight decay would provide a safeguard against overfitting. The default dropout parameters for each model [80] were retained.

When PyTorch dataloaders were initialized, a batch size of 64 was selected. Training dataloaders were set to shuffle the data to introduce randomness and help avoid overfitting, though for consistency shuffling was disabled for the validation and test dataloaders (in contrast to the TorchEEG documentation examples [80]).

The pipelines were run via Queen Mary University of London’s HPC facility [105].

4.4. Model evaluation

Eight models were tested, although as noted in section 5.2, three of these were impractical for use with DAAMEE-c and were not used to analyze it.

Given the large number of both models and data splitting techniques used, it was considered impractical to classify valence, arousal, and dominance separately, or to run eight-class VAD classification for all models. Additionally, based on previous research, it was assumed that patterns of performance would be at least broadly similar across all dimensions of emotion [47]. Thus, we chose to focus in the first instance on valence classification. This is similar to the approach taken by Zhang *et al*, who focused on classification of valence (and in some tests, arousal), when working with DEAP.

For a comprehensive analysis, the best-performing model on valence for a given dataset was used to classify arousal, dominance, and eight-class VAD combinations, using all splitting techniques. The best-performing model was judged as whichever algorithm achieved the highest mean accuracy across data splitting techniques for the dataset. This was considered a reasonable compromise, as it may not allow us to find the optimal feature extraction/model for each emotion dimension, but it does allow us to verify if similar performance is observed from the best valence classifier for each dataset when different dimensions of emotion are considered, and it allows us to test if the general trends (with regards to data splitting) are unique to valence.

To evaluate the overall performance from each test cycle, the test performance metrics were averaged over all folds (the number of the given dataset’s subjects for LOSO, $k = 10$ for all others). The mean and standard deviation (SD) for counts of training epochs were calculated similarly.

4.5. Statistical analyses

All three datasets have class imbalances, for all dimensions. Regarding DEAP, the most drastic single-class example is that 63% of trials have high dominance ratings. For DAAMEE, the worst single-class imbalance is that 57% of DAAMEE-c trials have high arousal ratings. Regarding eight-class classification, there are also imbalances in the label combinations for both datasets: for example, 31.25% of DEAP trials are high-valence, high-arousal, high-dominance, as opposed to 12.5% (balanced dataset). Chance accuracy is not 50% for an imbalanced dataset, so as a benchmark we consider the accuracy of a classifier that knows the class distribution and uses it to inform predictions. It can be shown (see appendix B) that for any random n -class classifier:

$$\text{Acc} = \sum_{i=1}^n p_i^2 \quad (1)$$

where p_i is the proportion of samples belonging to any class i . Binomial tests were conducted to evaluate whether classification performance was significantly

Table 1. Accuracies for classification of valence in the DEAP dataset, as percentages. The data splitting techniques are as follows: KFold (KF); KFoldGroupbyTrial (KFGbT); KFoldCrossTrial (KFCT); KFoldCrossSubject (KFCS); LeaveOneSubjectOut (LOSO) [77]. The models are as follows: the convolutional neural network (CNN) is the continuous CNN (CCNN) [79]; the graphical neural networks (GNNs) are the dynamical graph convolutional neural network (DGCNN) [106] and the local-global-graph network (LGGNet) [101]; the Transformers (Trans.) models are the basic, or ‘Vanilla’, Transformer, as well as the Vision Transformer (ViT) [107], and its derivatives, SimpleViT [108], and ArjunViT (developed by Arjun *et al*) [109]; and the recurrent neural network (RNN) is the long short-term memory (LSTM) network [26]. Bold values are significantly above-chance, i.e. above the $p < 0.05$ accuracy, which is 51.2%. Daggers (†) and double daggers (‡) indicate respectively the best and second-best performing models in each column.

Model type	Model	Split					Mean
		KF	KFGbT	KFCT	KFCS	LOSO	
CNN	CCNN	69.0[‡]	68.9[‡]	58.9[‡]	52.0	53.2	60.4[†]
GNN	DGCNN	69.4[†]	69.6[†]	58.7	51.5	51.6	60.2[‡]
	LGGNet	64.3	64.3	59.2[†]	50.6	51.8	58.0
Trans.	‘Vanilla’	58.6	58.6	55.2	55.7	54.5	56.5
	SimpleViT	56.6	56.6	56.6	56.7[†]	56.6[†]	56.6
	ViT	56.6	56.6	56.3	56.6[‡]	56.4[‡]	56.5
	ArjunViT	58.1	58.0	55.1	50.7	51.2	54.6
RNN	LSTM	64.0	63.7	58.7	48.7	49.0	56.8

greater ($p < 0.05$) than chance accuracy computed as in equation (1). Additionally, to verify if the variation in performance across different data splitting techniques is statistically significant, one-way ANOVA analysis was conducted for each dataset. For simplicity, mean accuracy values for each pipeline were used (as opposed to the accuracies from each fold).

5. Results

Section 5.1 gives descriptive statistics of the emotional ratings, both in the original dataset collected for the present work, and in DEAP. Section 5.2 reports the results for valence classification, with section 5.3 giving those for arousal, dominance, and eight-class VAD classification. Appendix C gives statistics on the training epochs completed for each pipeline.

5.1. Emotional responses

The experiment yielded valence, arousal and dominance ratings (0.00–10.00) for 16 stimuli from 31 subjects for a total sample size of 496, when considering all of the subjects, including those not in DAAMEE-c. Overall, the ratings show substantial variance in all three dimensions of emotion: valence (range: 9.85; mean: 5.40; SD: 1.92), arousal (range: 10.00; mean: 5.36; SD: 1.95), and dominance (range: 9.82; mean: 5.02; SD: 1.61). The overall mean rating for each dimension is within 0.50 of the scale midpoint, 5.00, indicating that there is not a substantial skew towards low or high ratings for valence, arousal, or dominance, for either of the datasets. An approximation across all three scales with mean = 5.00 and SD = 2.00 would suggest that roughly 68% (1 SD) of responses were between 3.00 and 7.00 with a further

27% (2 SDs) 1.00–2.00 or 8.00–9.00 and a further 5% (3 SDs) reaching the extremes of 0.00 and 10.00. This variance in the ratings is also reflected in individual subjects’ minimum and maximum ratings: subjects responded from an average minimum 2.87 to average maximum 7.94 (mean range = 5.06) for valence, 2.68–7.69 (mean range = 5.01) for arousal and 2.58–7.33 (mean range = 4.76) for dominance. Altogether, this high level of variance in emotion ratings provides a sound basis for emotion decoding from EEG.

5.2. Valence classification with all models

Tables 1–3 give the performance accuracies for valence classification with DEAP, DAAMEE-s, and DAAMEE-c respectively. The results show that for KF and KFGbT, the DGCNN was consistently the best performing model across all three datasets followed by the CCNN for the scalp EEG datasets (DEAP and DAAMEE-s) and the LGGNet for the cEEGGrid dataset (DAAMEE-c). For KFCT, there is much less differentiation between the models with accuracy varying from 55.1 to 59.2 across all datasets and data-splitting techniques. The two best-performing models are the LGGNet and the CCNN for DEAP, the RNN and the CCNN for DAAMEE-s, and the DGCNN and ‘Vanilla’ Transformer for DAAMEE-c. The results for KFCS and LOSO consistently show that the ‘Vanilla’ Transformer outperforms other models.

As noted in section 4.4, the best-performing model for each dataset on the valence decoding task was selected for use on the arousal, dominance and eight-class VAD decoding tasks. The best-performing model was judged to be the one that achieved the highest mean accuracy in valence classification across data splitting techniques. For DEAP valence classification, the highest mean accuracy

Table 2. Accuracies for classification of valence in DAAMEE-s, the scalp EEG subset of the original dataset collected for the present research, as percentages. The data splitting techniques are as follows: KFold (KF); KFoldGroupbyTrial (KFGbT); KFoldCrossTrial (KFCT); KFoldCrossSubject (KFCS); LeaveOneSubjectOut (LOSO) [77]. The models are as follows: the convolutional neural network (CNN) is the continuous CNN (CCNN) [79]; the graphical neural networks (GNNs) are the dynamical graph convolutional neural network (DGCNN) [106] and the local-global-graph network (LGGNet) [101]; the Transformers (Trans.) models are the basic, or ‘Vanilla’, Transformer, as well as the Vision Transformer (ViT) [107], and its derivatives, SimpleViT [108], and ArjunViT (developed by Arjun *et al*) [109]; and the recurrent neural network (RNN) is the long short-term memory (LSTM) network [26]. Bold values are significantly above-chance, i.e. above the $p < 0.05$ accuracy, which is 51.2%. Daggers (†) and double daggers (‡) indicate respectively the best and second-best performing models in each column.

Model type	Model	Split					Mean
		KF	KFGbT	KFCT	KFCS	LOSO	
CNN	CCNN	73.2[‡]	73.7[‡]	57.8[‡]	50.6	50.1	61.1[‡]
GNN	DGCNN	73.8[†]	74.0[†]	57.5	50.3	51.9	61.5[†]
	LGGNet	65.4	65.9	56.9	52.9	51.2	58.5
Trans.	‘Vanilla’	58.4	58.0	57.1	55.6[†]	54.8[†]	56.8
	SimpleViT	61.1	61.1	57.0	51.4	51.2	56.4
	ViT	59.8	59.5	56.4	50.5	54.2	56.1
	ArjunViT	61.4	61.3	56.3	53.1	54.6[‡]	57.4
RNN	LSTM	62.2	62.3	58.2[†]	55.5[‡]	54.6[‡]	58.6

Table 3. Accuracies for classification of valence in DAAMEE-c, the cEEGrid subset of the original dataset collected for the present research, as percentages. The data splitting techniques are as follows: KFold (KF); KFoldGroupbyTrial (KFGbT); KFoldCrossTrial (KFCT); KFoldCrossSubject (KFCS); LeaveOneSubjectOut (LOSO) [77]. Only some of the models tested in the present research are appropriate for use with cEEGrid data, and so were used in these tests. These models are as follows: the graphical neural networks (GNNs) are the dynamical graph convolutional neural network (DGCNN) [106] and the local-global-graph network (LGGNet) [101]; the Transformers (Trans.) models are the basic, or ‘Vanilla’, Transformer, as well as ArjunViT (developed by Arjun *et al*) [109], a derivative of the Vision Transformer (ViT) model [107]; and the recurrent neural network (RNN) is the long short-term memory (LSTM) network [26]. Bold values are significantly above-chance, i.e. above the $p < 0.05$ accuracy, which is 51.5%. Daggers (†) and double daggers (‡) indicate respectively the best and second-best performing models in each column.

Model type	Model	Split					Mean
		KF	KFGbT	KFCT	KFCS	LOSO	
GNN	DGCNN	75.0[†]	75.0[†]	59.2[†]	47.4	50.2	61.4[†]
	LGGNet	66.2[‡]	66.5[‡]	57.0	52.0	54.7[‡]	59.3[‡]
Trans.	‘Vanilla’	60.0	59.8	58.8[‡]	55.5[†]	58.0[†]	58.4
	ArjunViT	60.1	61.4	57.9	51.6	52.4	56.7
RNN	LSTM	61.9	61.0	56.8	52.3[‡]	51.0	56.6

achieved was 60.4%, when the CCNN was used. Notably, the dynamical graph convolutional neural network (DGCNN) model [106] performed nearly as well, achieving a mean accuracy of 60.2%, with all other mean accuracies for DEAP valence being $\leq 58.0\%$. For DAAMEE-s, again the CCNN and the DGCNN both out-performed the other models: the CCNN achieved 61.1% on average, and the DGCNN achieved 61.5%, making the DGCNN the best-performing model for DAAMEE-s. Finally, for DAAMEE-c valence classification there was a maximum mean accuracy of 61.4%, achieved with the DGCNN, making this the best-performing valence classifier for DAAMEE-c. Figure 5(a) displays the results for the best-performing classifiers. Figure 5

shows the reference DEAP accuracies. In that study the CCNN was the only model used for the data splitting benchmark test, however it was also found to be the best-performing model for DEAP classification in a different benchmark test [77].

As shown by the ANOVA test results in table 4, there were significant performance differences between data splitting techniques for each dataset. For all three datasets, KF and KFGbT are associated with higher accuracy, KFCT with intermediate accuracy, and KFCS and LOSO with lower accuracy (though still significantly greater than chance for some models). These differences are likely to reflect overfitting due to various forms of data leakage as discussed in section 6.

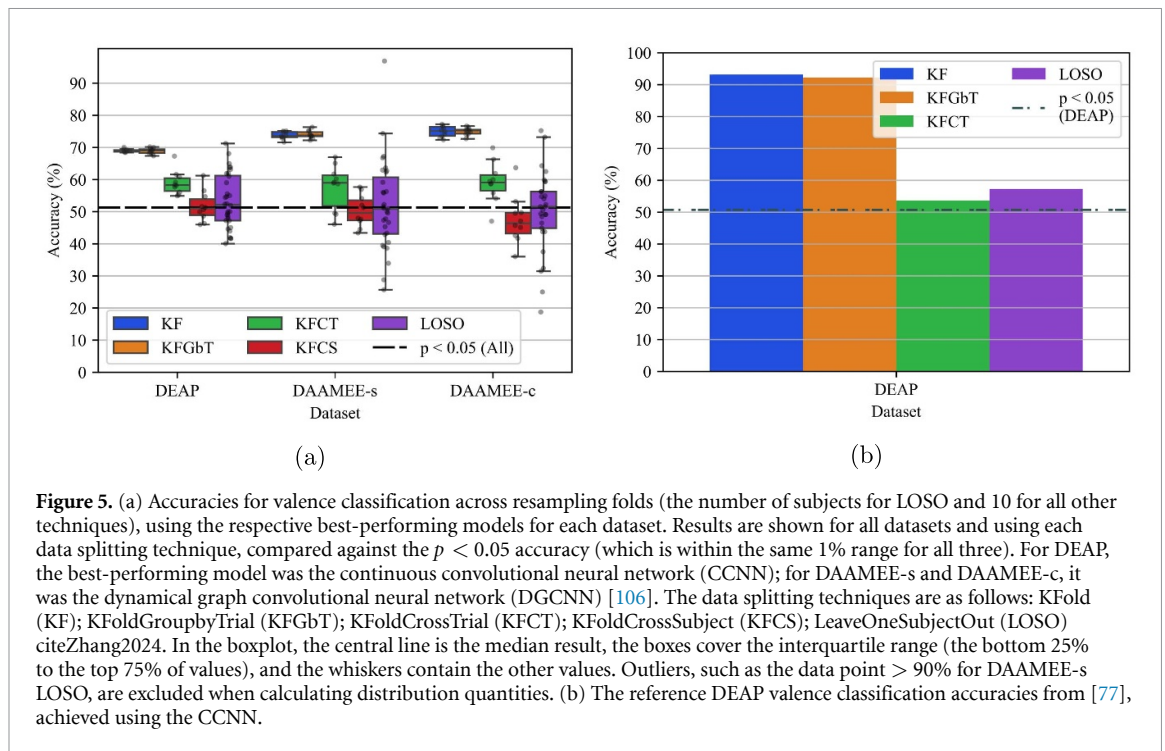


Figure 5. (a) Accuracies for valence classification across resampling folds (the number of subjects for LOSO and 10 for all other techniques), using the respective best-performing models for each dataset. Results are shown for all datasets and using each data splitting technique, compared against the $p < 0.05$ accuracy (which is within the same 1% range for all three). For DEAP, the best-performing model was the continuous convolutional neural network (CCNN); for DAAMEE-s and DAAMEE-c, it was the dynamical graph convolutional neural network (DGCNN) [106]. The data splitting techniques are as follows: KFold (KF); KFoldGroupbyTrial (KFGbT); KFoldCrossTrial (KFCT); KFoldCrossSubject (KFCS); LeaveOneSubjectOut (LOSO) citeZhang2024. In the boxplot, the central line is the median result, the boxes cover the interquartile range (the bottom 25% to the top 75% of values), and the whiskers contain the other values. Outliers, such as the data point $> 90\%$ for DAAMEE-s LOSO, are excluded when calculating distribution quantities. (b) The reference DEAP valence classification accuracies from [77], achieved using the CCNN.

Table 4. The results of the ANOVA tests for emotion decoding. Specifically, one-way ANOVA testing was performed to investigate the performance differences between data splitting techniques when each dataset was analyzed. F_{crit} values were found in [110]. DEAP, DAAMEE-s, and DAAMEE-c are the datasets, with the latter two being the scalp and cEEGrid subsets of the original dataset collected for the present research. Bold values are significantly above-chance ($p < 0.05$).

Dataset	F_{crit}	F_{Obs}	p
DEAP	2.64	10.9	7.72×10^{-6}
DAAMEE-s	2.64	16.9	8.18×10^{-8}
DAAMEE-c	2.87	9.48	1.81×10^{-4}

5.3. VAD classification with the best-performing models

Table 5 reports the performance accuracies for arousal, dominance, and eight-class VAD classification using the best-performing model for each dataset. Figure 6 presents these visually.

6. Discussion

6.1. Overview of results

Sections 6.2–6.4 discuss the results and their implications in detail. Nevertheless, it is useful to first consider the overall trends observed. The results in tables 1–5 show that the majority of models achieved significantly better than chance performance (this applies individually for all datasets and data splitting techniques, i.e. within each table column most accuracies are significantly above-chance). It should be noted however that while many pipelines yield accuracies which are significantly above-chance, they are still far from perfect performance. For example,

considering the results from DEAP valence classification shown in table 1, we can see that 35/40 classification pipelines yielded significantly above-chance performance, but only 8/40 achieved a performance accuracy of $\geq 60\%$, which is a fairly low threshold for a binary classifier to be considered reliable. 27/40 pipelines achieved between 51.2% (the $p < 0.05$ threshold for DEAP) and 60%. Broadly similar results are seen from the work classifying valence in DAAMEE-s and DAAMEE-c. These findings are notable in demonstrating that ear-EEG data analysis can yield emotion decoding performance similar to scalp EEG data analysis, despite the reduced spatial coverage of ear-EEG.

6.2. Data splitting

A key aim of this study was to investigate directly how the use of different data splitting techniques influences estimates of model performance, through the types of data leakage discussed in section 2.3. There was a focus on two particular mechanisms: temporal correlations within trials, and subject-specific properties.

6.2.1. Splitting technique and performance—overall trends

From inspecting the results, one can immediately see that models trained and tested using the KF and KFGbT data splits often perform substantially better than those trained and tested using KFCT, which in turn often outperform those tested using KFCS and LOSO. Both the KF and KFGbT techniques split the data in such a way that samples from a given trial and subject would be used for training, with data from

Table 5. Accuracies for classification of arousal (aro.), dominance (dom.), and eight-class valence-arousal-dominance (VAD) using each dataset's best-performing model (that which achieved peak valence classification accuracy), as percentages. The data splitting techniques are as follows: KFold (KF); KFoldGroupbyTrial (KFGbT); KFoldCrossTrial (KFCT); KFoldCrossSubject (KFCS); LeaveOneSubjectOut (LOSO) [77]. The models in this table are as follows: the convolutional neural network (CNN) is the continuous CNN (CCNN) [79]; the graphical neural network (GNN) is the Dynamic Graph CNN (DGCNN) [106]. DEAP, DAAMEE-s, and DAAMEE-c are the datasets, with the latter two being the scalp and cEEGrid subsets of the original dataset collected for the present research. Bold values are significantly above-chance ($p < 0.05$).

Dataset, model	Label	$p < 0.05$ acc.	Split					Mean
			KF	KFGbT	KFCT	KFCS	LOSO	
DEAP, CCNN(CNN)	Aro.	51.9	70.6	70.8	60.7	54.0	48.8	59.4
	Dom.	53.2	70.8	71.3	62.2	57.2	54.1	63.1
	VAD	17.9	50.1	50.8	30.2	20.0	21.2	34.5
DAAMEE-s, DGCNN(GNN)	Aro.	51.2	72.5	73.3	59.5	53.9	53.2	62.5
	Dom.	50.9	70.3	70.4	53.9	55.2	53.7	60.7
	VAD	16.1	48.2	50.0	25.7	21.3	20.9	33.2
DAAMEE-c, DGCNN(GNN)	Aro.	51.7	74.2	73.4	57.9	52.8	54.3	62.5
	Dom.	50.8	71.3	72.4	51.6	52.2	50.2	59.5
	VAD	16.2	49.2	49.1	24.5	16.1	18.4	31.5

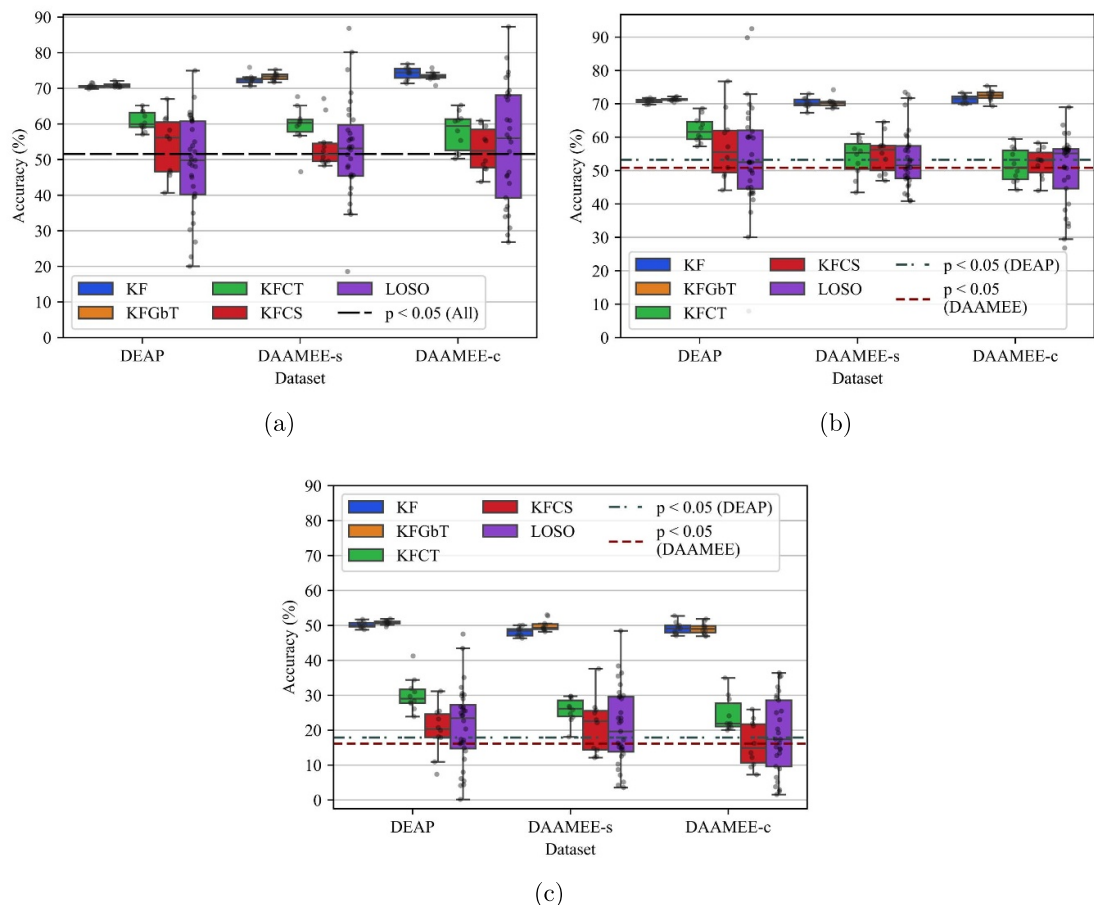


Figure 6. Accuracies across resampling folds (the number of subjects for LOSO and 10 for all other techniques) for each dataset and split technique for: (a) arousal classification; (b) dominance classification; and (c) eight-class valence-arousal-dominance (VAD) classification. The model used was the best performing model based on average valence decoding performance: For DEAP, this was the CCNN; for DAAMEE-s and DAAMEE-c, it was the DGCNN. The data splitting techniques are as follows: KFold (KF); KFoldGroupbyTrial (KFGbT); KFoldCrossTrial (KFCT); KFoldCrossSubject (KFCS); LeaveOneSubjectOut (LOSO) [77]. In the boxplot diagram, the middle line for each pipeline is the median result, the boxes cover the interquartile range (the bottom 25% to the top 75% of values), and the whiskers contain the other values. Outliers, such as the data points $>85\%$ for DEAP LOSO in (b), are excluded when calculating distribution quantities. $p < 0.05$ accuracies which are within a 1% range of each other have been averaged.

the same trial and subject also being used for testing (keeping in mind that the data were randomized during KF splitting). The KFCT technique splits the data such that samples from a given subject are used for both training and testing, with separation of trials between folds. The KFCS and LOSO techniques split the data such that different trials and subjects are separated between folds. Thus, the results show that data splitting techniques that allow the models to learn temporal correlations within trials and subject-specific properties of the EEG signal (KF and KFGbT) yield better performance estimates than those that allow exploitation of subject-specific properties only (KFCT) which yield better estimates than techniques that do not allow either of these forms of data leakage (KFCS, LOSO).

Exploitation of temporal correlations tends to have a greater effect than exploitation of subject-specific properties. For example, for the DGCNN working with DAAMEE-s the performance drop between KFGbT and KFCT was 16.5 percentage points, whereas the performance drop between KFCT and LOSO was only 5.6 percentage points. In other words, the performance drop when trial-based data leakage is prevented is greater than that when subject-based data leakage is also prevented (ignoring the effects of stimulus sharing for simplicity). Interestingly, this effect is not as strong for the weaker-performing models (such as the ‘Vanilla’ Transformer), suggesting that for these models temporal correlations in particular are not as effectively exploited. In any case, the fact that trial-based data leakage is often more powerful than subject-based data leakage indicates that this is of greater concern both in the present work and for biomedical machine learning research more generally.

An important limitation of the models that perform better for particular splitting techniques is that they are unlikely to generalize well to data recorded under different conditions (i.e. different subjects and recording sessions).

6.2.2. Subject-specific properties vs noise

Even when the data splitting technique does not allow exploitation of temporal correlations within trials, classification performance can be positively affected by subject- (and session-) specific properties of the EEG signal if data splitting techniques allow data from the same subject to be used both in training and testing, as discussed in section 2.3. Conversely, performance could be adversely affected by cross-subject noise consisting of aspects of the signal that make classification of data from other subjects more difficult (e.g. mood on the day). In the present context, the fact that KFCT and KFGbT extract an equal amount of data from each subject for each fold (with KF approximately doing so as well) may limit both positive effects of subject-specific properties and negative effects of cross-subject noise [111–114]. This

may partly explain why KF, KFGbT, and KFCT perform better than KFCS and LOSO which eliminate data leakage due to subject-specific properties but also increase the potential for cross-subject noise.

From the present results, we cannot make confident judgments on the extent to which the (relatively modest) performance differences between KFCT and KFCS/LOSO reflect exploitation of subject-specific properties boosting performance in KFCT, or negative effects of cross-subject noise in KFCS/LOSO. If the study were to be repeated, a possible improvement to better investigate this would be to conduct tests where systematic subject biases are enforced, for example, only using high-valence data for some subjects and low-valence data for others. However, to ensure that there is enough data for each class, this may necessitate using multi-dataset learning, or developing a new dataset in which different groups of subjects are exposed to stimuli designed to evoke different responses (having a low-valence and high-valence stimuli group, for example).

Finally, it is worth noting that there may be situations in which exploitation of subject-specific properties would not be considered data leakage and may even be beneficial. For example, in a single user application where a model is trained (or fine tuned) on data from the individual user, it would make sense to exploit subject-specific properties.

6.2.3. KF vs KFGbT and the impact of shuffling

By default (i.e. with no shuffling used), the KFGbT technique extracts data from a particular trial slice in each fold (e.g. the first 10% of each trial for test fold 0 with $k = 10$). This could potentially introduce a domain gap between the train and test data due to the test data covering a specific period of each trial which the model has not seen [77]. To aim for a fair comparison here we shuffled within trials for all k -fold techniques, not only KF as in [77], meaning that for each subject and trial the same amount of data will be taken, but from random periods. For example, when KFGbT is used to split DEAP without shuffling with $k = 10$, test fold 0 will contain the first 6 s of each trial, for each subject. When KFGbT is used to split DEAP with shuffling with $k = 10$, test fold 0 will contain six randomly picked 1 s periods from each trial, for each subject. This explains why in the previous benchmarks, pipelines featuring KFGbT (without shuffling) were consistently slightly outperformed by those featuring KF [77], but in our work, where there is not such a domain gap with KFGbT due to shuffling being used consistently, the performance results for KF and KFGbT are on par. In fact, when averaging all 27 binary classification results achieved using both splitting techniques, one finds a difference of <0.1 percentage points, indicating that when shuffling is used for both KF and KFGbT, neither leads to models that have an advantage over their counterparts.

6.2.4. Data leakage impacts different models differently

A major consideration for the interpretation of these results is that they are highly dependent on the model used. As an example, consider the valence classification results for DEAP (table 1). For the DGCNN model, the performance accuracies ranged from 52.3% to 69.9%. By contrast, for the SimpleViT model the performance accuracies ranged from 56.6% to 56.7%. This indicates that some models are better at exploiting temporal/subject-based features than others, thereby suffering more from bias due to data leakage.

6.2.5. Summary

The results demonstrate clear effects of KF and KFGbT on overfitting, through exploitation of temporal correlations, meaning that the trained models may not generalize well to new data recorded under different conditions. As a result, our recommendation for future research is to use other data splitting methods. For multi-user applications, where performance is expected to generalize across users, KFCS or LOSO (differing only in the number of folds) would be the preferred methods. For small datasets, LOSO might be preferred for reduced bias due to greater training set overlap between folds whereas for larger datasets KFCS provides a more tractable approach, reducing the amount of model training required for CV. For single-user applications, where networks can be trained or fine-tuned on the user's own data, KFCT is an acceptable alternative, allowing for exploitation of subject-specific properties of the neural signal.

6.3. Model performance

6.3.1. Best-performing models

The best-performing models, judged as those that achieved highest valence classification accuracy, averaged across all data splits, were the CCNN for DEAP, and the DGCNN for DAAMEE-s and DAAMEE-c. The CCNN also performed particularly well for DAAMEE-s, yielding a mean valence classification accuracy which was only 0.4 percentage points lower than the DGCNN value. Similarly, the DGCNN was shown to perform very well for DEAP classification, yielding a mean valence classification accuracy that was only 0.2 percentage points lower than the CCNN value. These results indicate that the CCNN and DGCNN are both effective classifiers for scalp EEG data, with the DGCNN also being effective for classifying cEEGrid data.

An important caveat to these results is that the criterion used may benefit models that are particularly affected by data leakage, as the mean accuracies will be skewed upwards by the results for data splitting techniques which allow data leakage. For example, when working with DAAMEE-c, the DGCNN performs especially well when the KF and KFGbT splits

are used (75.0% performance accuracy for both), particularly in contrast to splits such as KFCS and LOSO, for which it did not achieve statistically significant accuracy. The reason for this may be that the DGCNN is particularly effective at exploiting temporal correlations and subject-specific properties. As figure 3(b) shows, KF and KFGbT splitting allow data from shared trials and subjects to be used for the training/validation/test sets, whereas KFCS and LOSO do not allow data from either shared trials or common subjects to be used between sets. Thus, it is possible that the KF and KFGbT pipelines with the DGCNN may simply be very effective at exploiting data leakage, leading to the overall DGCNN mean accuracy being skewed upwards. By contrast, models such as the 'Vanilla' Transformer have lower mean accuracies but yield reasonably high performance across all data splitting techniques.

Aside from the question of data leakage, it is feasible that the feature extraction method is the dominant factor, as opposed to the performance of the models themselves: the CCNN and DGCNN's pipelines, and only these models' pipelines, involve the extraction of frequency band DE. Either this method is useful for both scalp and ear-EEG classification, or it emphasizes temporal correlations/subject-specific properties which are being exploited. The former interpretation would be in line with the literature [7], however the fact that performance is much poorer for the more stringent data splitting methods provides support for the latter interpretation. Notably, frequency band DE was shown to be a particularly useful method for music emotion decoding in the benchmark tests [77]; however since these utilized KF data splitting, the role of overfitting was not clear.

6.3.2. VAD classification with the best-performing models

For each dataset, the best-performing algorithm performed similarly when transitioning from valence classification to arousal/dominance classification. Proportionately, the performances were similar between datasets for the more difficult eight-class VAD classification. These trends indicate that, at least for the best-performing algorithms, the models are similarly effective for each of the classification tasks considered, and the trends regarding data splitting techniques are consistent for each task. A limited exception is that, as figure 6 shows, the performance drop from two-class to eight-class classification was not as large (proportionately or in absolute terms) when KF and KFGbT splitting were used. A possible explanation for this is that for more difficult problems, or when data for each class is less abundant, the models exploit temporal correlations to a greater extent.

6.3.3. Some models benefit more from cross-fold single subject data splitting

It is also interesting to note that for some models, using KFCT data splitting does not lead to better performance accuracy than using KFCS or LOSO splitting for otherwise identical pipelines, even if using KFGbT or KF data splitting does provide benefit. This has two possible implications: some models are effective at exploiting temporal correlations, but not subject-specific properties; or, these models are simply affected less by cross-subject noise. Considering the discussion in section 6.2, the latter explanation, or a combination of the two factors, seems most probable.

6.3.4. Compactness can benefit cEEGrid

Interestingly, the GNN models performed better overall when working with DAAMEE-c than with the scalp datasets. When averaging performance accuracies for the two GNNs (the DGCNN and the LGGNet) from classifying valence with each dataset, the DAAMEE-c value is the highest (60.2%, compared to 59.1% for DEAP and 60.0% for DAAMEE-s). As discussed in section 6.3.1, a caveat to this is that the results may have been skewed due to data leakage, which is supported by the fact that the results for the more stringent data splitting techniques are substantially lower than for the others. However, the results for the Transformers (the ‘Vanilla’ Transformer and ArjunViT) also show better performance for valence classification with DAAMEE-c compared to the scalp datasets (a mean accuracy of 57.6% for DAAMEE-c, compared to 55.6% for DEAP, and 57.1% for DAAMEE-s). Furthermore, the DAAMEE-c analysis pipelines with these models yielded similar results across all splits, indicating that data leakage effects were less likely than for the GNNs. These results suggest that certain model types may be more effective at working with cEEGrid data than they are at working with scalp data, even when accounting for possible data leakage effects. A plausible explanation for this is that these models benefit from cEEGrid having fewer channels and being more spatially limited than 32-channel scalp EEG, as the lower dimensionality of the input data may make it easier for models to prioritize information that is useful for classification. This would be consistent with the literature, where it has been found that the temporal lobe scalp locations are among the most relevant for EEG emotion decoding [39]. In fact, Choi *et al* showed that ear-EEG may be better-suited to subject-independent emotion decoding than scalp EEG for this reason [7].

6.3.5. Better learners are better cheaters

It must be emphasized that the data splitting technique has a substantial impact on performance, such that where some models learn better than others, they

may primarily be learning to exploit features such as temporal correlations due to data leakage. When considering only the KFCT results, a different model (the LGGNet) performs best for valence classification in the DEAP dataset. When considering only the KFCS and LOSO valence classification results, the SimpleViT performs best for DEAP, whereas the ‘Vanilla’ Transformer performs best for DAAMEE-s and DAAMEE-c.

6.4. Datasets

As discussed in section 6.1, most pipelines yielded significantly above-chance classification performance for all datasets. Before comparing pipeline performance across datasets, it is useful to compare chance performance across them. The somewhat similar levels of class imbalance, as well as each dataset being split into many sub-trial epochs for classification (which improves the confidence of statistical predictions), led to the datasets having similar $p < 0.05$ threshold values. For valence classification these are all within 0.3 percentage points, for arousal classification these are all within 0.7 percentage points, for dominance classification these are all within 2.5 percentage points (with DEAP being slightly less balanced than the other two datasets, increasing the chance accuracy), and for VAD classification these are within 1.8 percentage points. This consistency facilitates comparisons across the datasets and classification tasks/emotion dimensions.

The results indicate that the DAAMEE dataset (including both the scalp data in DAAMEE-s and the cEEGrid data in DAAMEE-c) is of reasonable quality. Not only were most accuracies achieved during DAAMEE classification significantly above-chance performance, but they were generally similar to those achieved during DEAP classification. Although data leakage of the forms discussed in section 2.3 will have inflated some of the performance accuracies, some of the splits, particularly KFCS and LOSO, prevent these types of leakage; even for these splits, the majority of classification pipelines yield significantly above-chance performance accuracies with the DAAMEE data. The fact that performance accuracies were so similar to those of the tried-and-tested DEAP dataset for the same task strongly suggests that the models are able to extract and use similar amounts of useful information from each dataset, indicating that the datasets are of similar quality. The results also suggest similar emotion decoding performance between cEEGrid and scalp EEG.

DAAMEE differs from DEAP in several respects. First, the stimulus consist of music only rather than music video. Therefore, the results demonstrate emotion decoding in purely musical listening, which is useful since real-world musical listening often consists of isolated audio without corresponding video.

Second, the music was specially composed which has advantages in terms of experimental control, ensuring that the stimuli were unfamiliar to all subjects, but also means that the results may not generalize to listening of familiar music. On the other hand, it would be reasonable to suppose that emotional effects would be stronger for familiar music, so there is no reason to believe the results would not generalize. Third, the stimuli were limited to three instruments which has advantages in terms of experimental control since emotion decoding cannot be confounded by timbral differences between stimuli. On the other hand, the results might not generalize to music with different instrumental textures, though it is worth noting that a range of different musical styles were represented in the stimulus set.

6.5. Limitations

6.5.1. Sources of noise

There is inherent noise in any empirical measure of subjective emotional experience. Several steps were taken to mitigate this: explaining the VAD scales via text, speech and diagrams; encouraging subjects to use the full range of the scale; and binarizing the ratings into low and high categories before the analysis. Some recordings suffered from drift noise. To address this, we focused on ensuring robust electrode connections, and used a 1 Hz highpass filter and a linear detrending algorithm in the preprocessing sequence. In the cEEGrid setup used, shown in figure 2(a), flexible silicone arrays were attached to the head using double-sided stickers, instead of a widely-used setup featuring rigid arrays [115]. Our setup facilitated more individualized electrode fittings than using rigid arrays, but a drawback was that it could be difficult to attach electrodes. cEEGrid data from one subject were excluded from analysis as an insufficient number of decent connections were made.

6.5.2. Limited data per subject

Each subject completed one practice trial and 15 main trials, each 30 s long. This gives 8 min of data per subject in total, or 7 min 34.72 s for the analyses conducted here. This is relatively little data, and we have used the practice trials in the analysis as the need to maximize the amount of usable data was thought to outweigh the potential drawbacks. This lack of data per subject is compensated to an extent by the relatively large sample of subjects (29 for DAAMEE-c, 31 for DAAMEE-s, as discussed in section 3.2). Cross-subject analysis was conducted, so generally the overall amount of data is more important than the data for each subject; for the LOSO data split technique, only one subject was used for each of the validation and test sets in each split, so having relatively few trials per subject may have been more problematic for this split. Even so, given the large number of splits (one per subject), the average variance would have been

reduced. Were the experiment to be repeated, possible improvements would be to conduct more trials and to make trials longer (perhaps 1 min long, similar to those in DEAP). Subject fatigue was a concern during experiment planning, but this could be reduced by encouraging the subject to take longer breaks, or, if necessary, by collecting each subject's data over two sessions.

6.6. Future work

In the previous subsections, we have noted specific limitations to both this study and the DAAMEE dataset, and how these may be overcome if the work were repeated. There are also several ways that the analysis could be built upon in follow-up research.

6.6.1. Data splitting with a consideration of stimuli

It remains possible that KFCS and LOSO derive their above-chance performance from the fact that data relating to the same stimuli but different subjects are included in the training/validation/test sets (see section 2.3 and figure 3(b)). For DAAMEE, this is mitigated by three factors: (1) subjects were split into two groups who experienced different stimuli, and within each group, different subjects could be assigned different practice stimuli (one out of 16 trials); (2) the stimulus presentation order was randomized for each subject; (3) the emotional experiences differed between subjects, meaning that the same stimuli were assigned different emotion labels by them. With regards to DEAP, the same 40 stimuli are used for each subject, albeit with the order randomized for each subject. This means that point (1) is not applicable, and so the issue is slightly worse for this dataset. In any case, future research should examine data splitting techniques that prevent both subjects and stimuli from overlapping between training and test sets. Such techniques could include LOPMO [72], or the fold-based approach used by Nguyen *et al* [70].

6.6.2. Feature extraction techniques

The investigation of feature extraction techniques could be augmented in two ways: by taking a more systematic approach, using all feature extraction techniques in combination with all models (at least, where compatible); and by using a wider range of features, such as approximate entropy, which the previous benchmarks have shown to be a slightly more useful feature for the CCNN than band DE [77].

6.6.3. Hyperparameter tuning

As well as allowing us to raise the performance of classifiers generally, implementing hyperparameter tuning may allow for fairer comparisons between models, feature extraction techniques, and data splitting techniques, by avoiding the assumption that using the default/example parameters affects all pipelines

equally. Care would have to be taken to avoid additional overfitting.

6.6.4. Shallow classifiers

Another addition could be to use shallow classifiers in the analysis; these are not currently in the TorchEEG library, but could be otherwise implemented. This would be particularly astute as shallow classifiers have been shown to be less prone to overfitting [116–119], including that caused by within-trial temporal correlations [57].

6.6.5. Statistical analysis

Binomial tests were used to compare decoding accuracy of the trained classifiers with chance performance. These tests assume independence for the individual measurements from which the accuracy score is computed (i.e. matches between predicted emotional ratings and actual ratings). However, where the splitting technique allows for data leakage between training and test sets (i.e. KF, KFGbT and, to a lesser extent, KFCT), the assumption of independence will not entirely hold. Since both matching and non-matching trials would be equally affected, this should not produce systematic bias of predictive accuracy but may reduce the effective sample size and, therefore, underestimate the actual variance. This could mean that greater differences from chance performance are required for significance. One approach is to use data splitting methods such as LOSO or KFCS that avoid data leakage in the first place, as we have argued above for reasons of robust generalization. This would reduce the risk of generating correlated outputs but would not rule it out entirely. Therefore, in future research this would be best addressed by testing directly for distorted variance through a comparison of the actual variance in accuracy across trials with the expected variance from the binomial distribution. If the actual variance is higher than the expected variance, the effective sample size should be adjusted accordingly in the binomial tests.

6.6.6. More comprehensive VAD classification study

A more comprehensive study of VAD classification might provide further insight into model performance. Using only the best-performing valence classifier for each dataset to classify arousal, dominance, and eight-class VAD was justified given the large number of models featured in this study, but a compromise in future studies may be to use the best two or three performing classifiers for the three dimensions of emotion.

7. Conclusion

This work investigated emotion decoding using EEG data, with a focus on the mobile cEEG technique, and the impact of the data splitting technique used.

We collected a novel dataset, DAAMEE, including both scalp and ear-EEG data from an emotion-based task where subjects listened to music and recorded their emotions. Analyses compared model performance on DAAMEE with the existing DEAP dataset. In DEAP, subjects watched music videos, whereas DAAMEE is more specific in reflecting musically-induced emotions only.

The results show above-chance performance for valence, arousal, dominance and eight-class VAD classification for all datasets. For the cEEGGrid data, performance is comparable to the corresponding scalp EEG data and the DEAP dataset. As an example, the DGCNN model working with DAAMEE-c classified valence with an overall mean classification accuracy of 61.4%, whilst the same model working with DEAP and DAAMEE-s achieved average accuracies of in the ranges of 60.2% and 61.5%.

The results also indicate that emotion classifier performance can be influenced by the data splitting technique used. This is exemplified by the results for VAD classification with DAAMEE-c, where using two of the data splitting techniques (including the commonly-used random k -fold) led to accuracies of $\geq 49.1\%$, compared to the $p < 0.05$ baseline accuracy of 16.2%, and values of $\leq 24.5\%$ when other techniques were used. For some techniques, including random k -fold, performance is likely to be inflated unless constraints are implemented to avoid indirect data leakage based on temporal correlations and subject-specific properties. The results show this in the context of emotion classification, but these trends can be assumed to be universal (consistent with the literature). When conducting EEG machine learning research, it is crucial to carefully consider, evaluate and report the data splitting paradigm. It is also important to consider the requirements of any given practical application. For example, in real-life EEG emotion decoding, it may be useful to allow subject-specific features to be learned if training is conducted with the user's own data.

DAAMEE provides a new dataset for benchmarking music emotion decoding models from scalp and ear-EEG data. Further work could include: an extension of the investigation into data splitting focusing on stimulus-based data leakage; and a more expansive analysis of model performance, for example with a more detailed investigation of feature extraction techniques and hyperparameter optimization.

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Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: <https://doi.org/10.5281/zenodo.15798525> [120].

Ethics statement

The research was conducted in accordance with the principles embodied in the Declaration of Helsinki and with local statutory requirements. All participants gave written informed consent to participate in the study. The research was given prior approval by the Aarhus University Research Ethics Committee (Institutional Review Board), approval number TECH 2023–006.

Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial

relationships that could be construed as a potential conflict of interest.

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Supplementary materials

The code (such as PsychoPy scripts) and stimuli which were used to run the experiment are available at: <https://doi.org/10.5281/zenodo.16038975>.

The code used for the analysis is available at: <https://doi.org/10.5281/zenodo.16038785>.

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Appendix A. Code and technical details for analysis

Table A.1 gives the software used in the analysis. Table A.2 gives details of the preprocessing and hyperparameters used in the machine learning pipelines (with the exception of the DEAP-CCNN-Alt extension; see appendix D). Please see the TorchEEG documentation for further information [80].

Table A.1. Software details. DAAMEE-s and DAAMEE-c are the scalp and cEEGrid subsets of the original dataset collected for this research.

Software	Version	Notes
Python	3.12.7	–
MNE-Python	1.6.1	Used for DAAMEE-s and DAAMEE-c preprocessing.
PyTorch	2.5.1	–
PyTorch lightning	2.5.0.post0	–
TorchEEG	1.1.2	Error observed in <code>transforms/numpy</code> files <code>band.py</code> and <code>band_pyeeg.py</code> . The command: <code>from scipy.signal import butter, lfilter, hann</code> must be corrected to: <code>from scipy.signal import butter, lfilter from scipy.signal.window import hann</code> This has been fixed in newer versions.

Table A.2. Transforms and hyperparameters used for the convolutional neural network (CNN) and graphical neural network (GNN) models, for both the DEAP and DAAMEE datasets, including the DAAMEE-c subdataset. Abbreviations are as follows: continuous CNN (CCNN) [79]; dynamic graph convolutional neural network (DGCNN) [106]; and local-global-graph network (LGGNet) [101]. Default values are only noted where they are used for DAAMEE and not for DEAP. In addition to the transforms listed, all of which are offline, **ToTensor** is applied as an online transform for each pipeline. Transforms are applied in the orders given. The table is continued for the Transformer models and the recurrent neural network (RNN) on the next page. Transforms and hyperparameters used for the Transformer (Trans.) and recurrent neural network (RNN) models, for both the DEAP and DAAMEE datasets, including the DAAMEE-c subdataset. Abbreviation meanings are as follows: Transformer (Trans.); Vision Transformer (ViT) [107], with SimpleViT [108], and ArjunViT (developed by Arjun *et al*) [109], being ViT derivatives; and long-short term memory (LSTM) [26]. Default values are only noted where they are used for DAAMEE and not for DEAP. In addition to the transforms listed, all of which are offline, **ToTensor** is applied as an online transform for each pipeline. Transforms are applied in the orders given. The first half of the table, on the previous page, contains details for the CCNN and the GNN models.

Model	Transforms and non-default parameters	Non-default classifier hyperparameters	Used for DAAMEE-c	DAAMEE adaptations
CNN (CCNN)	- BandDifferentialEntropy. - ToGrid.	—	No	- BandDifferentialEntropy band_dict = {'delta': (1, 4), 'theta': (4, 8), 'alpha': (8, 13), 'beta': (13, 30), 'gamma': (30, 45)}. - in_channels = 5 input to classifier.
DCGNN (GNN)	BandDifferentialEntropy.	- in_channels = 4. - num_electrodes = 32.	Yes	- BandDifferentialEntropy band_dict = {'delta': (1, 4), 'theta': (4, 8), 'alpha': (8, 13), 'beta': (13, 30), 'gamma': (30, 45)}. - in_channels = 5 input to classifier.
LGGNet (GNN)	- MeanStdNormalize. - To2d.	DEAP_GENERAL_ REGION_LIST (from torcheeg. datasets. constants. emotion_recognition .deap) input.	Yes	- MeanStdNormalizenot implemented. - DAAMEE-s region list created, using format_ region_channel_list, GENERAL_REGION_LIST from the torcheeg.datasets.constants utils andregion_1020 classes respectively. - DAAMEE-c region list created, with one channel per region. - chunk_size = 203 input to classifier. - num_electrodes = 20 input to classifier for DAAMEE-c.
‘Vanilla’ (Trans.)	—	head_channels= 64.	Yes	- head_channels = 8 (default),chunk_size = 203, t_patch_size = 29 (an integer divisor of the chunk size) input to classifier. - num_electrodes = 20 input to classifier for DAAMEE-c.
SimpleViT (Trans.)	- MinMaxNormalize(axis = -1). - ToGrid(DEAP_CHANNEL_ LOCATION_DICT).	—	No	- MinMaxNormalizenot implemented. - Bespoke grid (of same size) created for DAAMEE-s, using 10–20 system coordinates [89]. - chunk_size = 203, t_patch_size = 29 (an integer divisor of the chunk size) input to classifier.

(Continued.)

Table A.2. (Continued.)

Model	Transforms and non-default parameters	Non-default classifier hyperparameters	Used for DAAMEE-c	DAAMEE adaptations
ViT (Trans.)	• <code>MinMaxNormalize(axis = -1)</code>. • <code>ToGrid(DEAP_CHANNEL_LOCATION_DICT)</code>.	—	No	• <code>MinMaxNormalize</code> not implemented. • Bespoke grid (of same size) created for DAAMEE-s, using 10–20 system coordinates [89]. • <code>chunk_size = 203, t_patch_size = 29</code> (an integer divisor of the chunk size) input to classifier.
ArjunViT (Trans.)	—	—	Yes	• <code>chunk_size = 203, t_patch_size = 29</code> (an integer divisor of the chunk size) input to classifier • <code>num_electrodes = 20</code> input to classifier for DAAMEE-c.
LSTM (RNN)	—	—	Yes	• LSTM <code>num_electrodes = 20</code> for DAAMEE-c.

Appendix B. Adjusted chance performance derivation

Section 4.5 gives the accuracy equation for a random n -class classifier. The following is a derivation of those equations.

Let us first consider accuracy in the case of a binary classifier ($n = 2$), specifically a random classifier which knows the prior distribution and uses this to inform predictions. It will give correct predictions when it predicts that a sample belongs to class 1, and the sample is indeed in class 1, or when it predicts class 2 for a sample that is in class 2. Hence, the accuracy is given by (B.1).

$$\begin{aligned} \text{Acc} = & p(\text{prediction} = 1)p(\text{groundtruth} = 1) \\ & + p(\text{prediction} = 2)p(\text{groundtruth} = 2). \end{aligned} \quad (\text{B.1})$$

As $p(\text{prediction} = 1)$ is the proportion of samples which are in class 1, which is equivalent to $p(\text{groundtruth} = 1)$, we can substitute $p(\text{prediction} = 1) \equiv p(\text{groundtruth} = 1) \equiv p_1$, and likewise for class 2:

$$\text{Acc}_{\text{chance}} = p_1^2 + p_2^2. \quad (\text{B.2})$$

This reasoning can be extended to any random n -class classifier:

$$\text{Acc} = \sum_{i=1}^n p_i^2. \quad (\text{B.3})$$

That is (1) in the main paper.

Appendix C. Training epoch data

Tables C.1–C.3 give statistics on the training epochs completed for valence classification in DEAP, DAAMEE-s, and DAAMEE-c respectively. These relate to the number of epochs completed before either early stopping occurs, or the epoch limit of 50 is reached, when each pipeline is run.

Appendix D. DEAP-CCNN-Alt

Table D.1 gives details of the preprocessing and hyperparameters used in the DEAP-CCNN-Alt pipelines, in which baseline removal techniques were applied to the preprocessed DEAP dataset (as advised in some pages of the TorchEEG documentation [80] and the TorchEEG code comments). Note that these tests were not run with DAAMEE (either DAAMEE-s or DAAMEE-c), as baseline periods were not recorded for trials in this dataset. Table D.2 gives the performance accuracies for valence classification. Table D.3 gives statistics on the training epochs completed.

Table C.1. Mean ($\bar{E}$) and standard deviation (σ_E) training epochs completed when each pipeline is applied to valence classification in the DEAP dataset. The data splitting techniques are as follows: KFold (KF); KFoldGroupbyTrial (KFGbT); KFoldCrossTrial (KFCT); KFoldCrossSubject (KFCS); LeaveOneSubjectOut (LOSO) [77]. The models are as follows: the convolutional neural network (CNN) is the continuous CNN (CCNN) [79]; the graphical neural networks (GNNs) are the dynamical graph convolutional neural network (DGCNN) [106] and the local-global-graph network (LGGNet) [101]; the Transformers (Trans.) models are the basic, or ‘Vanilla’, Transformer, as well as the Vision Transformer (ViT) [107], and its derivatives, SimpleViT [108], and ArjunViT (developed by Arjun *et al*) [109]; and the recurrent neural network (RNN) is the long short-term memory (LSTM) network [26].

Model type	Model	Split									
		KF		KFGbT		KFCT		KFCS		LOSO	
		$\bar{e}$	σ_e	$\bar{e}$	σ_e	$\bar{e}$	σ_e	$\bar{e}$	σ_e	$\bar{e}$	σ_e
CNN	CCNN	29.4	4.4	29.5	4.8	17.6	5.8	19.40	8.31	15.6	5.2
GNN	DGCNN	49.6	1.2	49.3	1.6	20.7	8.4	21.2	10.9	17.3	9.9
	LGGNet	32.0	5.1	33.2	4.4	20.8	8.2	14.1	2.2	18.5	9.7
Trans.	‘Vanilla’	37.9	8.2	40.1	9.4	19.0	9.4	18.4	8.5	19.1	10.5
	SimpleViT	11.0	0.0	11.0	0.0	11.1	0.3	11.0	0.0	11.0	0.0
	ViT	11.0	0.0	11.0	0.0	11.0	0.0	12.4	4.2	11.3	1.7
	ArjunViT	15.5	2.9	14.8	2.8	16.4	5.4	13.4	3.4	17.1	8.5
RNN	LSTM	34.4	7.5	29.0	4.9	21.3	7.4	13.5	5.4	15.3	7.7

Table C.2. Mean ($\bar{E}$) and standard deviation (σ_E) training epochs completed when each pipeline is applied to valence classification in DAAMEE-s, the scalp EEG subset of the original dataset collected for the present research. The data splitting techniques are as follows: KFold (KF); KFoldGroupbyTrial (KFGbT); KFoldCrossTrial (KFCT); KFoldCrossSubject (KFCS); LeaveOneSubjectOut (LOSO) [77]. The models are as follows: the convolutional neural network (CNN) is the continuous CNN (CCNN) [79]; the graphical neural networks (GNNs) are the dynamical graph convolutional neural network (DGCNN) [106] and the local-global-graph network (LGGNet) [101]; the Transformers (Trans.) models are the basic, or ‘Vanilla’, Transformer, as well as the Vision Transformer (ViT) [107], and its derivatives, SimpleViT [108], and ArjunViT (developed by Arjun *et al*) [109]; and the recurrent neural network (RNN) is the long short-term memory (LSTM) network [26].

Model type	Model	Split									
		KF		KFGbT		KFCT		KFCS		LOSO	
		$\bar{e}$	σ_e	$\bar{e}$	σ_e	$\bar{e}$	σ_e	$\bar{e}$	σ_e	$\bar{e}$	σ_e
CNN	CCNN	30.8	5.7	28.4	8.7	16.9	5.8	14.2	3.1	16.1	6.1
GNN	DGCNN	49.2	1.8	48.4	2.8	21.2	9.4	17.0	9.6	17.5	7.0
	LGGNet	27.9	9.3	27.6	8.9	17.4	4.2	12.4	1.4	15.3	4.4
Trans.	‘Vanilla’	19.1	5.0	19.6	7.0	20.9	8.1	18.4	7.7	15.7	5.4
	SimpleViT	28.7	8.9	26.5	4.8	19.6	9.5	20.7	11.3	19.0	10.1
	ViT	24.0	7.0	30.8	5.6	19.4	8.0	13.9	5.1	17.0	8.7
	ArjunViT	21.1	5.0	20.2	4.2	17.0	6.5	16.2	6.2	16.4	9.0
RNN	LSTM	31.7	11.2	31.3	13.1	27.0	12.3	14.2	4.6	18.7	8.8

Table C.3. Mean ($\bar{E}$) and standard deviation (σ_E) training epochs completed when each (cEEGrid-compatible) pipeline is applied to valence classification in DAAMEE-c, the cEEGrid subset of the original dataset collected for the present research. The data splitting techniques are as follows: KFold (KF); KFoldGroupbyTrial (KFGbT); KFoldCrossTrial (KFCT); KFoldCrossSubject (KFCS); LeaveOneSubjectOut (LOSO) [77]. Only some of the models tested in the present research are appropriate for use with cEEGrid data, and so were used in these tests. These models are as follows: the graphical neural networks (GNNs) are the dynamical graph convolutional neural network (DGCNN) [106] and the local-global-graph network (LGGNet) [101]; the Transformers (Trans.) models are the basic, or ‘Vanilla’, Transformer, as well as ArjunViT (developed by Arjun *et al*) [109], a derivative of the Vision Transformer (ViT) model [107]; and the recurrent neural network (RNN) is the long short-term memory (LSTM) network [26].

Model type	Model	Split									
		KF		KFGbT		KFCT		KFCS		LOSO	
		$\bar{e}$	σ_e	$\bar{e}$	σ_e	$\bar{e}$	σ_e	$\bar{e}$	σ_e	$\bar{e}$	σ_e
GNN	DGCNN	49.9	0.3	48.0	4.0	20.9	12.4	17.1	6.0	17.4	8.9
	LGGNet	2.0	3.0	2.0	3.0	2.0	3.0	1.0	0.00	1.8	4.4
Trans.	‘Vanilla’	19.0	10.7	14.8	2.5	16.7	6.8	15.7	7.3	14.7	8.4
	ArjunViT	21.5	7.2	21.6	7.5	13.4	3.6	11.9	0.9	17.6	10.1
RNN	LSTM	33.0	13.2	29.2	12.4	18.8	12.0	15.8	8.3	19.3	10.7

Table C.4. Mean ($\bar{E}$) and standard deviation (σ_E) training epochs completed for each dataset's best-performing model (that which achieved peak valence classification accuracy) when applied to arousal, dominance, and eight-class valence-arousal-dominance (VAD) classification. The data splitting techniques are as follows: KFold (KF); KFoldGroupbyTrial (KFGbT); KFoldCrossTrial (KFCT); KFoldCrossSubject (KFCS); LeaveOneSubjectOut (LOSO) [77]. The models are as follows. The convolutional neural network (CNN) is the continuous CNN (CCNN) [79]; the graphical neural network (GNN) is the dynamical graph convolutional neural network (DGCNN) [106].

Dataset	Model	Label	Split									
			KF		KFGbT		KFCT		KFCS		LOSO	
			$\bar{e}$	σ_e	$\bar{e}$	σ_e	$\bar{e}$	σ_e	$\bar{e}$	σ_e	$\bar{e}$	σ_e
DEAP	CCNN (CNN)	Aro.	33.4	3.1	31.9	3.4	14.4	3.9	15.2	3.0	16.6	6.9
		Dom.	33.4	3.0	31.8	2.4	15.7	5.1	15.7	4.2	14.0	6.6
DAAMEE-s	DGCNN (GNN)	VAD	40.4	3.4	40.0	3.7	13.0	2.5	11.6	0.8	12.7	3.7
		Aro.	44.7	8.5	47.8	6.6	19.0	5.9	21.3	12.2	19.1	8.4
		Dom.	49.7	0.9	49.2	2.4	19.1	8.0	22.5	11.2	21.9	11.0
DAAMEE-c	DGCNN (GNN)	VAD	50.0	0.0	50.0	0.0	19.6	9.9	14.7	3.1	15.4	5.1
		Aro.	49.3	1.8	49.5	1.5	21.7	10.8	13.2	3.1	15.2	4.4
		Dom.	49.0	3.0	50.0	0.0	15.4	3.9	19.0	10.7	20.3	9.1
		VAD	50.0	0.0	50.0	0.0	17.2	7.6	17.2	10.4	13.6	4.9

Table D.1. Transforms and hyperparameters used for DEAP-CCNN-Alt. Default values are excluded here. All transforms except for `baselineRemoval` are offline. Additionally, `ToTensor` is applied as an online transform for each pipeline. Transforms are applied in the orders given.

Transforms and non-default parameters	Non-default classifier hyperparameters
• <code>BandDifferentialEntropy(apply_to_baseline = True)</code>. • <code>ToGrid(apply_to_baseline = True)</code>. • <code>BaselineRemoval</code> (online, before <code>ToTensor</code>).	—

Table D.2. Accuracies for the DEAP-CCNN-Alt pipelines classifying valence, as percentages. The data splitting techniques are as follows: KFold (KF); KFoldGroupbyTrial (KFGbT); KFoldCrossTrial (KFCT); KFoldCrossSubject (KFCS); LeaveOneSubjectOut (LOSO) [77]. Bold values are significantly above chance ($p < 0.05$).

Split				
KF	KFGbT	KFCT	KFCS	LOSO
93.8	94.1	54.3	53.2	52.5

Table D.3. Mean ($\bar{E}$) and standard deviation (σ_E) training epochs completed when DEAP-CCNN-Alt is applied to valence classification. The data splitting techniques are as follows: KFold (KF); KFoldGroupbyTrial (KFGbT); KFoldCrossTrial (KFCT); KFoldCrossSubject (KFCS); LeaveOneSubjectOut (LOSO) [77].

Split									
KF		KFGbT		KFCT		KFCS		LOSO	
$\bar{e}$	σ_e	$\bar{e}$	σ_e	$\bar{e}$	σ_e	$\bar{e}$	σ_e	$\bar{e}$	σ_e
44.6	6.2	46.0	5.0	14.9	4.1	16.8	6.8	18.2	8.6

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